



Top 93 CAT Functions, Graphs and Statistics Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

Suppose for all integers x , there are two functions f and g such that $f(x) + f(x-1) - 1 = 0$ and $g(x) = x^2$. If $f(x^2 - x) = 5$, then the value of the sum $f(g(5)) + g(f(5))$ is

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Suppose for all integers x , there are two functions f and g such that $f(x) + f(x-1) - 1 = 0$ and $g(x) = x^2$. If $f(x^2 - x) = 5$, then the value of the sum $f(g(5)) + g(f(5))$ is

Answer: $f(25) + [f(5)]^2$

$g(5) = 25$

$x = 0$
 $f(0) = 5$
 $f(1) + f(0) = 1$
 $f(1) = -4$
 $f(2) = 1$
 $f(2) + f(1) = 1$

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Question 2

Consider two sets $A = \{2, 3, 5, 7, 11, 13\}$ and $B = \{1, 8, 27\}$. Let f be a function from A to B such that for every element in B , there is at least one element a in A such that $f(a) = b$. Then, the total number of such functions f is

- A 665
- B 667
- C 537
- D 540

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A) 665
C) 537

B) 667
D) 540

$f(2) = 1$
 $f(3) = 8$
 $f(5) = 1$
 $f(7) = 27$
 $f(11) = 1$
 $f(13) = 8$

$f(2) = 1$
 $f(3) = 8$
 $f(5) = 1$
 $f(7) = 27$
 $f(11) = 1$
 $f(13) = 8$

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Question 3

The number of distinct real values of x , satisfying the equation $\max\{x, 2\} - \min\{x, 2\} = |x + 2| - |x - 2|$, is

The number of distinct real values of x , satisfying the equation $\max\{x, 2\} - \min\{x, 2\} = |x + 2| - |x - 2|$, is

Answer: _____

$x \leq -2$
 $2 - x = -x - 2$
 $2 = -2$
 $x \geq 2$
 $2 - x = x - 2$
 $2 = x - 2$
 $x = 4$

$-2 < x < 2$

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Question 4

Find the sum $\sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} + \sqrt{1 + \frac{1}{2^2} + \frac{1}{3^2}} + \dots + \sqrt{1 + \frac{1}{2007^2} + \frac{1}{2008^2}}$

- A $2008 - \frac{1}{2008}$
- B $2007 - \frac{1}{2007}$
- C $2007 - \frac{1}{2008}$
- D $2008 - \frac{1}{2007}$
- E $2008 - \frac{1}{2009}$

Find the sum $\sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} + \sqrt{1 + \frac{1}{2^2} + \frac{1}{3^2}} + \dots + \sqrt{1 + \frac{1}{2007^2} + \frac{1}{2008^2}}$

A) $2008 - \frac{1}{2008}$
 B) $2007 - \frac{1}{2007}$
 C) $2007 - \frac{1}{2008}$
 D) $2008 - \frac{1}{2007}$

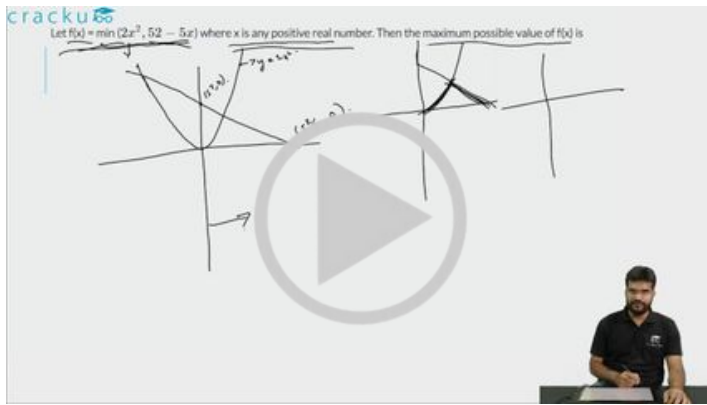
$\sqrt{1 + \frac{1}{a^2} + \frac{1}{(a+1)^2}} = \frac{a^2(a+1)^2 + (a+1)^2 + a^2}{a^2(a+1)^2}$
 $= \frac{a^2(a^2 + 2a + 1) + (a+1)^2 + a^2}{a^2(a+1)^2}$
 $= \frac{a^4 + 2a^3 + a^2 + a^2 + 2a + 1 + a^2}{a^2(a+1)^2}$
 $= \frac{a^4 + 2a^3 + 3a^2 + 2a + 1}{a^2(a+1)^2}$
 $= \frac{(a^2 + a + 1)^2}{a^2(a+1)^2}$
 $= \frac{a^2 + a + 1}{a(a+1)}$

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Question 5

Let $f(x) = \min(2x^2, 52 - 5x)$ where x is any positive real number. Then the maximum possible value of $f(x)$ is

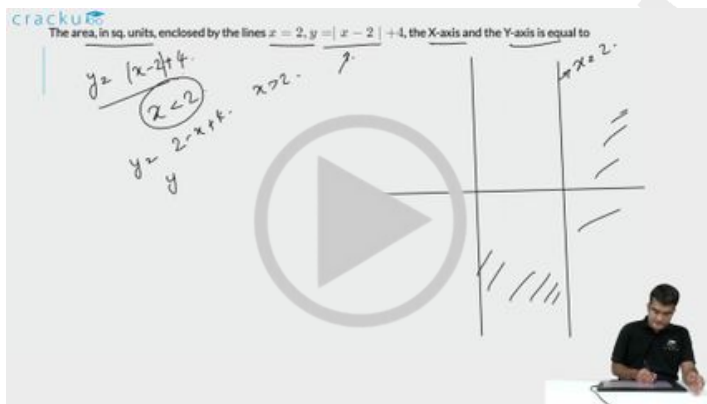


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Question 6

The area, in sq. units, enclosed by the lines $x = 2$, $y = |x - 2| + 4$, the X-axis and the Y-axis is equal to

- A 10
- B 6
- C 8
- D 12



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Question 7

Let $f(x) = \max(2x + 1, 3 - 4x)$, where x is any real number. Then the minimum possible value of $f(x)$ is:

- A $1/3$
- B $1/2$
- C $2/3$

D $4/3$

E $5/3$

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Question 8

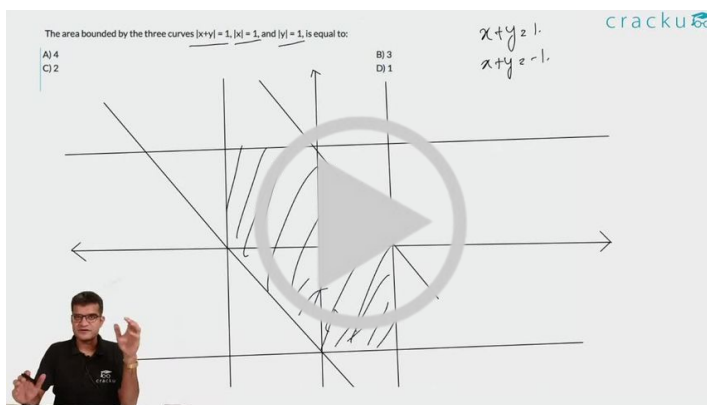
The area bounded by the three curves $|x+y| = 1$, $|x| = 1$, and $|y| = 1$, is equal to:

A 4

B 3

C 2

D 1



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Question 9

For all non-negative integers x and y , $f(x, y)$ is defined as below:

$$f(0, y) = y + 1$$

$$f(x+1, 0) = f(x, 1)$$

$$f(x+1, y+1) = f(x, f(x+1, y))$$

Then, what is the value of $f(1, 2)$?

A Two

B Four

C Three

D Cannot be determined

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Question 10

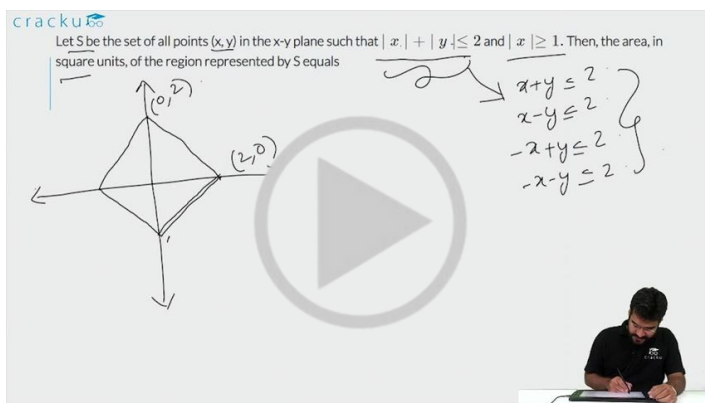
Let $Y = \text{minimum of } \{(x+2), (3-x)\}$. What is the maximum value of Y for $0 \leq x \leq 1$?

- A 1.0
- B 1.5
- C 3.1
- D 2.5

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Question 11

Let S be the set of all points (x, y) in the x - y plane such that $|x| + |y| \leq 2$ and $|x| \geq 1$. Then, the area, in square units, of the region represented by S equals



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Question 12

If $y = f(x)$ and $f(x) = \frac{(1-x)}{(1+x)}$, which of the following is true?

- A $f(2x) = f(x) - 1$
- B $x = f(2y) - 1$
- C $f(1/x) = f(x)$
- D $x = f(y)$





CAT Syllabus PDF



Question 13

What is the value of k for which the following system of equations has no solution:
 $2x - 8y = 3$ and $kx + 4y = 10$

- A -2
- B 1
- C -1
- D 2

What is the value of k for which the following system of equations has no solution:
 $2x - 8y = 3$ and $kx + 4y = 10$

AI-2
 CI-1

MO 1
 DI 2

$\rightarrow a_1x + b_1y = c_1$
 $a_2x + b_2y = c_2$

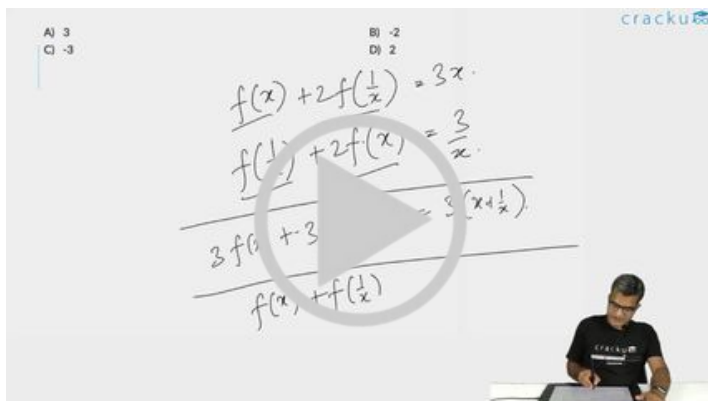
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \rightarrow \text{unique}$
 $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \rightarrow \text{no solution}$
 $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \rightarrow \text{no solution}$

$2x \rightarrow 2x + 4y = 10$
 $x + 6y = 20$

Question 14

For any non-zero real number x , let $f(x) + 2f\left(\frac{1}{x}\right) = 3x$. Then, the sum of all possible values of x for which $f(x) = 3$, is

- A 3
- B -2
- C -3
- D 2



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Question 15

If $\frac{a}{b+c} = \frac{b}{a+c} = \frac{c}{b+a} = r$, then r cannot take any value except

- A $1/2$
- B -1
- C $1/2$ or -1
- D $-1/2$ or -1



VIDEO SOLUTION



Question 16

A function can sometimes reflect on itself, i.e. if $y = f(x)$, then $x = f(y)$. Both of them retain the same structure and form. Which of the following functions has this property?

- A $y = \frac{2x+3}{3x+4}$
- B $y = \frac{2x+3}{3x-2}$
- C $y = \frac{3x+4}{4x-5}$

D None of the above

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Question 17

If $f(x+2) = f(x) + f(x+1)$ for all positive integers x , and $f(11) = 91$, $f(15) = 617$, then $f(10)$ equals

Handwritten solution for Question 17:

If $f(x+2) = f(x) + f(x+1)$ for all positive integers x , and $f(11) = 91$, $f(15) = 617$, then $f(10)$ equals

$f(10) = b$

$f(11) = 91$

$f(12) = f(10) + f(11)$

$= b + 91$

$f(13) = (b + 91) + 91 = 182 + b$

$f(14)$

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Question 18

For two positive integers a and b define the function $h(a,b)$ as the greatest common factor (G.C.F) of a, b . Let A be a set of n positive integers. $G(A)$, the GCF of the elements of set A is computed by repeatedly using the function h .

The minimum number of times h is required to be used to compute G is:

- A $1/2 n$
- B $(n - 1)$
- C n
- D None of these

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Question 19

x	1	2	3	4	5	6
y	4	8	14	22	32	44

In the above table, for suitably chosen constants a, b and c , which one of the following best describes the relation between y and x ?

- A $y = a + bx$
- B $y = a + bx + cx^2$

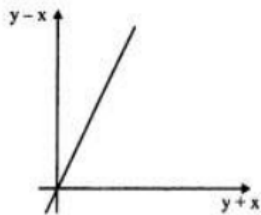
C $y = e^{a+bx}$

D None of the above

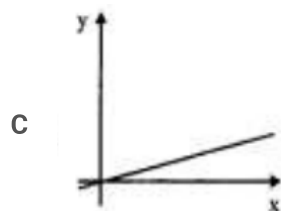
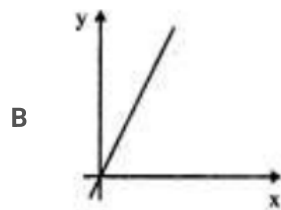
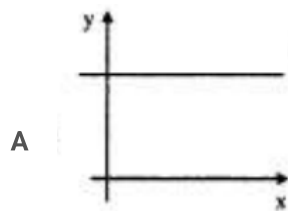
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Question 20

The graph of $y - x$ (on the y axis) against $y + x$ (on the x axis) is as shown below. (All graphs in this question are drawn to scale and the same scale and the same scale has been used on each axis.)



Which of the following shows the graph of y against x ?



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Question 21

Let $f(x) = 2x - 5$ and $g(x) = 7 - 2x$. Then $|f(x) + g(x)| = |f(x)| + |g(x)|$ if and only if

A $\frac{5}{2} < x < \frac{7}{2}$

B $x \leq \frac{5}{2}$ or $x \geq \frac{7}{2}$

C $x < \frac{5}{2}$ or $x \geq \frac{7}{2}$

D $\frac{5}{2} \leq x \leq \frac{7}{2}$

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Let $f(x) = 2x - 5$ and $g(x) = 7 - 2x$. Then $|f(x) + g(x)| = |f(x)| + |g(x)|$ if and only if

a. $5/2 < x < 7/2$
 b. $x < 5/2$ or $x \geq 7/2$
 c. $x \leq 5/2$ or $x \geq 7/2$
 d. $7/2 \leq x \leq 5/2$

① $f(x) > 0, g(x) < 0$

$f(x) - g(x)$

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CAT Percentile Predictor

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Question 22

The area of the quadrilateral bounded by the Y-axis, the line $x = 5$, and the lines $|x - y| - |x - 5| = 2$, is

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The area of the quadrilateral bounded by the Y-axis, the line $x = 5$, and the lines $|x - y| - |x - 5| = 2$, is

Answer: _____

$|x - 5| = 5 - x$
 $|x - y| - 5 + x = 2$
 $|x - y| = 7 - x$
 $\frac{y - x}{y - x} = \frac{7 - x}{7 - x}$
 $y = 7$

VIDEO SOLUTION

Question 23

The set of all positive integers is the union of two disjoint subsets:

$\{f(1), f(2), \dots, f(n), \dots\}$ and $\{g(1), g(2), \dots, g(n), \dots\}$, where $f(1) < f(2) < \dots < f(n) < \dots$, and $g(1) < g(2) < \dots < g(n) < \dots$, and $g(n) = f(f(n)) + 1$ for all $n \geq 1$. What is the value of $g(1)$?

- A Zero
- B Two
- C One
- D Cannot be determined

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Question 24

Suppose, the seed of any positive integer n is defined as follows:

$\text{seed}(n) = n$, if $n < 10$

$\text{seed}(n) = \text{seed}(s(n))$, otherwise, where $s(n)$ indicates the sum of digits of n .

For example, $\text{seed}(7) = 7$,

$\text{seed}(248) = \text{seed}(2 + 4 + 8) = \text{seed}(14) = \text{seed}(1 + 4) = \text{seed}(5) = 5$ etc.

How many positive integers n , such that $n < 500$, will have $\text{seed}(n) = 9$?

- A 39
- B 72
- C 81
- D 108
- E 55

Suppose, the seed of any positive integer n is defined as follows:
 $\text{seed}(n) = n$, if $n < 10$
 $\text{seed}(n) = \text{seed}(s(n))$, otherwise, where $s(n)$ indicates the sum of digits of n .
For example, $\text{seed}(7) = 7$,
 $\text{seed}(248) = \text{seed}(2 + 4 + 8) = \text{seed}(14) = \text{seed}(1 + 4) = \text{seed}(5) = 5$ etc.
How many positive integers n , such that $n < 500$, will have $\text{seed}(n) = 9$?

A) 39
B) 72
C) 81
D) 108
E) 55

Handwritten note: $468, 18$ n is a multiple of 9. $\text{seed}(n) = 9$.

[VIDEO SOLUTION](#)

CAT Score Calculator

Question 25

If $f(x) = (x^2 + 3x)(x^2 + 3x + 2)$ then the sum of all real roots of the equation $\sqrt{f(x) + 1} = 9701$, is

- A -6
- B 6
- C 3
- D -3

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Question 26

For real values of x , the range of the function $f(x) = \frac{2x-3}{2x^2+4x-6}$ is

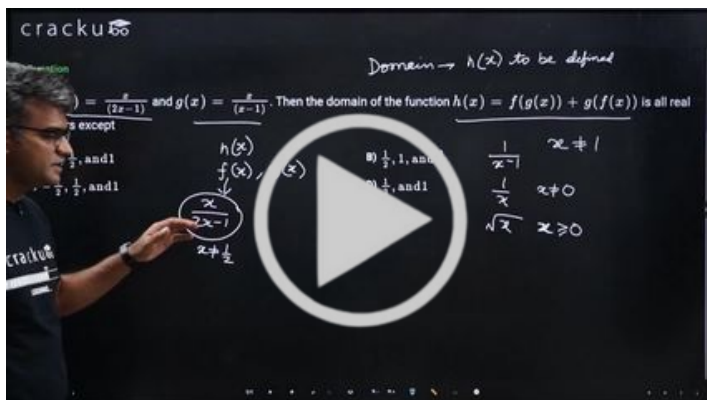
- A $(-\infty, \frac{1}{8}] \cup [1, \infty)$
- B $(-\infty, \frac{1}{4}] \cup [1, \infty)$
- C $(-\infty, \frac{1}{8}] \cup [\frac{1}{2}, \infty)$
- D $(-\infty, \frac{1}{4}] \cup [\frac{1}{2}, \infty)$

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Question 27

Let $f(x) = \frac{x}{(2x-1)}$ and $g(x) = \frac{x}{(x-1)}$. Then the domain of the function $h(x) = f(g(x)) + g(f(x))$ is all real numbers except

- A $-1, \frac{1}{2}$, and 1
- B $\frac{1}{2}, 1$, and $\frac{3}{2}$
- C $-\frac{1}{2}, \frac{1}{2}$, and 1
- D $\frac{1}{2}$, and 1



[VIDEO SOLUTION](#)



Question 28

$f(x) = \frac{5x+2}{3x-5}$ and $g(x) = x^2 - 2x - 1$, then the value of $g(f(f(3)))$ is

- A 2
- B $\frac{1}{3}$
- C 6
- D $\frac{2}{3}$

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If $f(x) = (5x + 2)/(3x - 5)$ and $g(x) = x^2 - 2x - 1$, then find out the value of $g(f(f(3)))$.

$f(3) = 17/4$
 $f(f(3)) = f(17/4) = \dots$

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Question 29

The number of the real roots of the equation $2 \cos(x(x + 1)) = 2^x + 2^{-x}$ is

- A 2
- B 1
- C infinite
- D 0

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The number of the real roots of the equation $2 \cos(x(x + 1)) = 2^x + 2^{-x}$ is

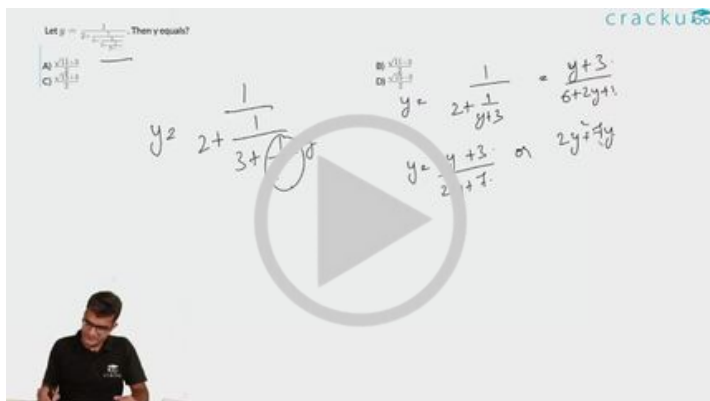
$2 \cos(x(x + 1)) \leq 2$
 $2^x + 2^{-x} \geq 2$
 $\left[2^x + \frac{1}{2^x} \right] \geq 2$
 $= 2$ when $2^x = 1$
 $x = 0$

VIDEO SOLUTION

Question 30

Let $y = \frac{1}{2 + \frac{1}{3 + \frac{1}{2 + \frac{1}{3 + \dots}}}}$. Then y equals?

- A $\frac{\sqrt{11}+3}{2}$
- B $\frac{\sqrt{11}-3}{2}$
- C $\frac{\sqrt{15}+3}{2}$
- D $\frac{\sqrt{15}-3}{2}$



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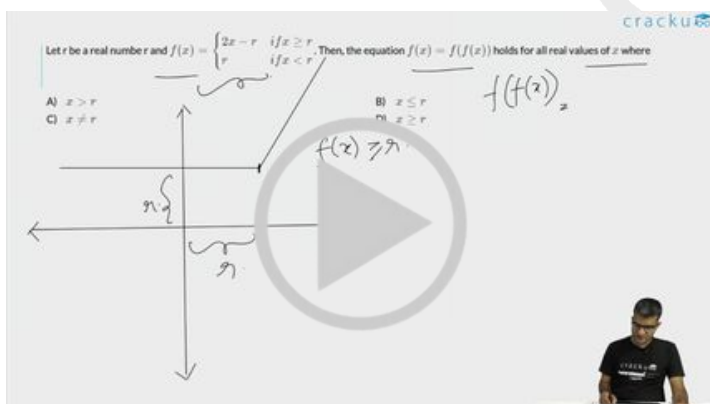
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Question 31

Let r be a real number and $f(x) = \begin{cases} 2x - r & \text{if } x \geq r \\ r & \text{if } x < r \end{cases}$. Then, the equation $f(x) = f(f(x))$ holds for all real values of x where

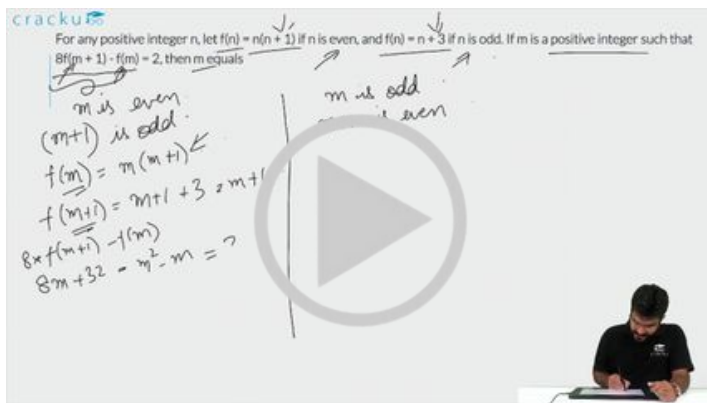
- A $x > r$
- B $x \leq r$
- C $x \neq r$
- D $x \geq r$



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Question 32

For any positive integer n , let $f(n) = n(n + 1)$ if n is even, and $f(n) = n + 3$ if n is odd. If m is a positive integer such that $8f(m + 1) - f(m) = 2$, then m equals

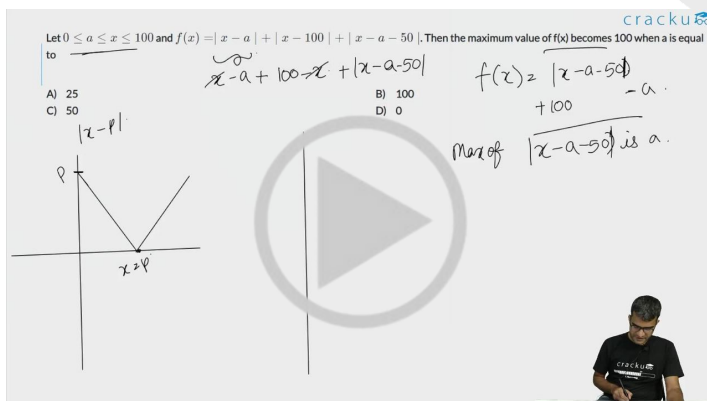


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Question 33

Let $0 \leq a \leq x \leq 100$ and $f(x) = |x - a| + |x - 100| + |x - a - 50|$. Then the maximum value of $f(x)$ becomes 100 when a is equal to

- A 25
- B 100
- C 50
- D 0



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Question 34

For any real number x , let $[x]$ be the largest integer less than or equal to x . If $\sum_{n=1}^N \left[\frac{1}{5} + \frac{n}{25} \right] = 25$ then N is

For any real number x , let $[x]$ be the largest integer less than or equal to x . If $\sum_{n=1}^N \left[\frac{1}{5} + \frac{n}{25} \right] = 25$ then N is

Answer: $\left[\frac{1}{5} + \frac{1}{25} \right] = 0$, $\left[\frac{1}{5} + \frac{2}{25} \right] = 0$, $\left[\frac{1}{5} + \frac{3}{25} \right] = 0$, $\left[\frac{1}{5} + \frac{4}{25} \right] = 0$, $\left[\frac{1}{5} + \frac{5}{25} \right] = 1$, $\left[\frac{1}{5} + \frac{6}{25} \right] = 1$, $\left[\frac{1}{5} + \frac{7}{25} \right] = 1$, $\left[\frac{1}{5} + \frac{8}{25} \right] = 1$, $\left[\frac{1}{5} + \frac{9}{25} \right] = 1$, $\left[\frac{1}{5} + \frac{10}{25} \right] = 2$, $\left[\frac{1}{5} + \frac{11}{25} \right] = 2$, $\left[\frac{1}{5} + \frac{12}{25} \right] = 2$, $\left[\frac{1}{5} + \frac{13}{25} \right] = 2$, $\left[\frac{1}{5} + \frac{14}{25} \right] = 2$, $\left[\frac{1}{5} + \frac{15}{25} \right] = 3$, $\left[\frac{1}{5} + \frac{16}{25} \right] = 3$, $\left[\frac{1}{5} + \frac{17}{25} \right] = 3$, $\left[\frac{1}{5} + \frac{18}{25} \right] = 3$, $\left[\frac{1}{5} + \frac{19}{25} \right] = 3$, $\left[\frac{1}{5} + \frac{20}{25} \right] = 4$, $\left[\frac{1}{5} + \frac{21}{25} \right] = 4$, $\left[\frac{1}{5} + \frac{22}{25} \right] = 4$, $\left[\frac{1}{5} + \frac{23}{25} \right] = 4$, $\left[\frac{1}{5} + \frac{24}{25} \right] = 5$.

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Question 35

Let $f(x) = \max(5x, 52 - 2x^2)$, where x is any positive real number. Then the minimum possible value of $f(x)$

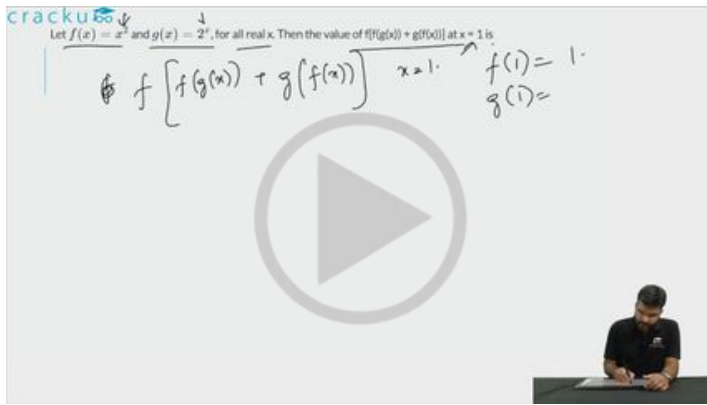
Let $f(x) = \max(5x, 52 - 2x^2)$, where x is any positive real number. Then the minimum possible value of $f(x)$

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Question 36

Let $f(x) = x^2$ and $g(x) = 2^x$, for all real x . Then the value of $f[f(g(x)) + g(f(x))]$ at $x = 1$ is

- A 16
- B 18
- C 36
- D 40



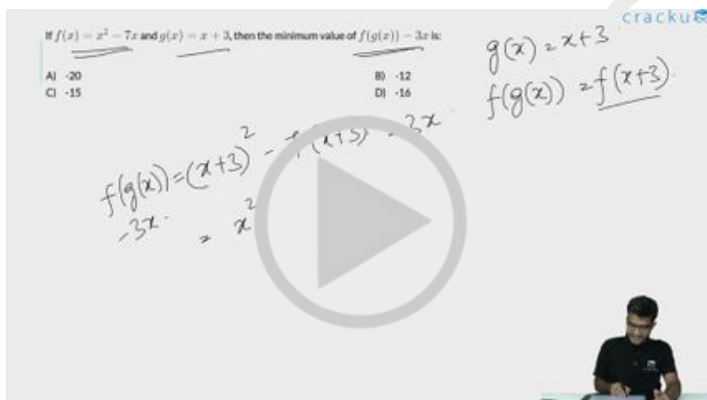
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Question 37

If $f(x) = x^2 - 7x$ and $g(x) = x + 3$, then the minimum value of $f(g(x)) - 3x$ is:

- A -20
- B -12
- C -15
- D -16



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Question 38

A function $f(x)$ satisfies $f(1) = 3600$, and $f(1) + f(2) + \dots + f(n) = n^2 f(n)$, for all positive integers $n > 1$. What is the value of $f(9)$?

- A 80
- B 240
- C 200

D 100

E 120

A function $f(x)$ satisfies $f(1) = 3600$, and $f(1) + f(2) + \dots + f(n) = n^2 f(n)$, for all positive integers $n > 1$. What is the value of $f(9)$?

A) 80
B) 240
C) 200
D) 100
E) 120

The handwritten solution shows the derivation of $f(n) = \frac{n}{n+2} f(n+1)$ from the given condition. It starts with $f(1) + f(2) + \dots + f(n) = n^2 f(n)$ and $f(1) + f(2) + \dots + f(n+1) = (n+1)^2 f(n+1)$. Subtracting the first equation from the second gives $f(n+1) = (n+1)^2 f(n+1) - n^2 f(n)$, which simplifies to $f(n+1) = \frac{n}{n+2} f(n+1)$.

VIDEO SOLUTION

Question 39

Let f be a function such that $f(mn) = f(m)f(n)$ for every positive integers m and n . If $f(1)$, $f(2)$ and $f(3)$ are positive integers, $f(1) < f(2)$, and $f(24) = 54$, then $f(18)$ equals

Let f be a function such that $f(mn) = f(m)f(n)$ for every positive integers m and n . If $f(1)$, $f(2)$ and $f(3)$ are positive integers, $f(1) < f(2)$, and $f(24) = 54$, then $f(18)$ equals

The handwritten solution starts with $f(1) > 1$, $f(2) = f(2 \cdot 1) = f(2)f(1)$, and $f(3) = f(3 \cdot 1) = f(3)f(1)$. It then shows $f(24) = 54$ and $f(18) = ?$. It also notes $f(1) < f(2)$ and $f(1) < f(3)$.

VIDEO SOLUTION

CAT Study Material

All CAT Free Resources

Question 40

If $f(x) = x^3 - 4x + p$, and $f(0)$ and $f(1)$ are of opposite signs, then which of the following is necessarily true [CAT 2004]

- A $-1 < p < 2$
- B $0 < p < 3$
- C $-2 < p < 1$
- D $-3 < p < 0$

If $f(x) = x^3 - 4x + p$, and $f(0)$ and $f(1)$ are of opposite signs, then which of the following is necessarily true

[CAT 2004]

A) $-1 < p < 2$
 B) $0 < p < 3$
 C) $-2 < p < 1$
 D) $-3 < p < 0$

$f(0) \times f(1) < 0$



VIDEO SOLUTION

Question 41

Let $f(x) \neq 0$ for any 'x' be a function satisfying $f(x)f(y) = f(xy)$ for all real x, y. If $f(2) = 4$, then what is the value of $f(\frac{1}{2})$?

- A 0
- B $\frac{1}{4}$
- C $\frac{1}{2}$
- D 1
- E cannot be determined

Let $f(x) \neq 0$ for any 'x' be a function satisfying $f(x)f(y) = f(xy)$ for all real x, y. If $f(2) = 4$, then what is the value of $f(\frac{1}{2})$?

A) 0
 B) $\frac{1}{4}$
 C) $\frac{1}{2}$
 D) 1
 E) cannot be determined

$x=y=1$
 $f(1)f(1) = f(1)$



VIDEO SOLUTION

Question 42

Let $f(x) = ax^2 + bx + c$, where a, b and c are certain constants and $a \neq 0$?

It is known that $f(5) = -3f(2)$. and that 3 is a root of $f(x) = 0$.

What is the other root of $f(x) = 0$?

- A -7
- B -4
- C 2

D 6

E cannot be determined

Let $f(x) = ax^2 + bx + c$, where a, b and c are certain constants and $a \neq 0$?
It is known that $f(5) = -3f(2)$, and that 3 is a root of $f(x) = 0$.
What is the other root of $f(x) = 0$?
[CAT 2008]

A) -7
B) 4
C) 2
E) cannot be determined

$f(x) = a(x-3)(x-p)$
 $f(5) = a \times 2 \times (5-p)$
 $f(2) =$

VIDEO SOLUTION



Question 43

Let $f(x) = ax^2 + bx + c$, where a, b and c are certain constants and $a \neq 0$?

It is known that $f(5) = -3f(2)$, and that 3 is a root of $f(x) = 0$.

What is the value of $a + b + c$?

A 9

B 14

C 13

D 37

E cannot be determined

Let $f(x) = ax^2 + bx + c$, where a, b and c are certain constants and $a \neq 0$?
It is known that $f(5) = -3f(2)$, and that 3 is a root of $f(x) = 0$.
What is the value of $a + b + c$?
[CAT 2008]

A) 9
B) 14
C) 13
D) 37
E) cannot be determined

$f(x) = 0$
 $f(3) = 0$
 $9a + 3b + c = 0 \quad \text{--- (1)}$
 $25a + 5b + c = -3[4a + 2b + c]$

VIDEO SOLUTION

Question 44

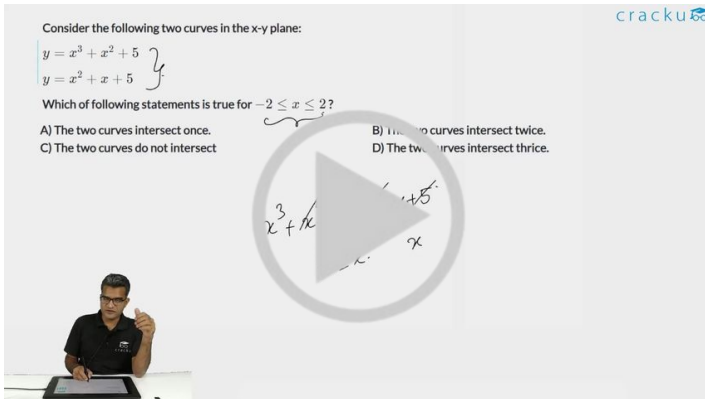
Consider the following two curves in the x-y plane:

$$y = x^3 + x^2 + 5$$

$$y = x^2 + x + 5$$

Which of following statements is true for $-2 \leq x \leq 2$?

- A The two curves intersect once.
- B The two curves intersect twice.
- C The two curves do not intersect
- D The two curves intersect thrice.

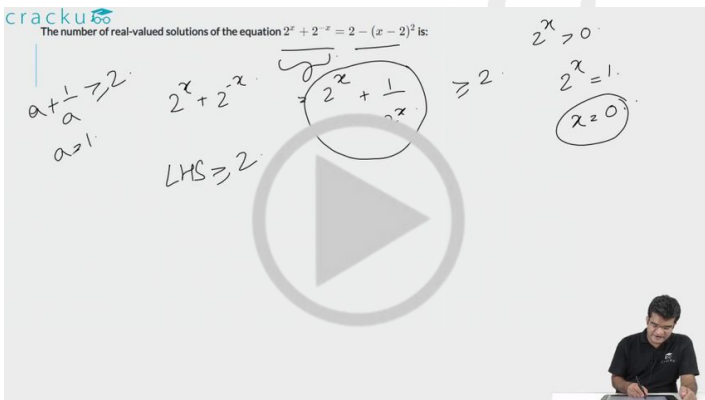


VIDEO SOLUTION

Question 45

The number of real-valued solutions of the equation $2^x + 2^{-x} = 2 - (x - 2)^2$ is:

- A 1
- B 2
- C infinite
- D 0



VIDEO SOLUTION

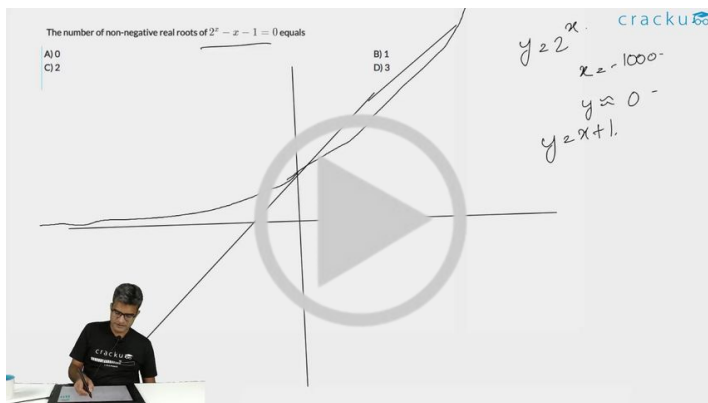
100+ CAT Quant Questions



Question 46

The number of non-negative real roots of $2^x - x - 1 = 0$ equals

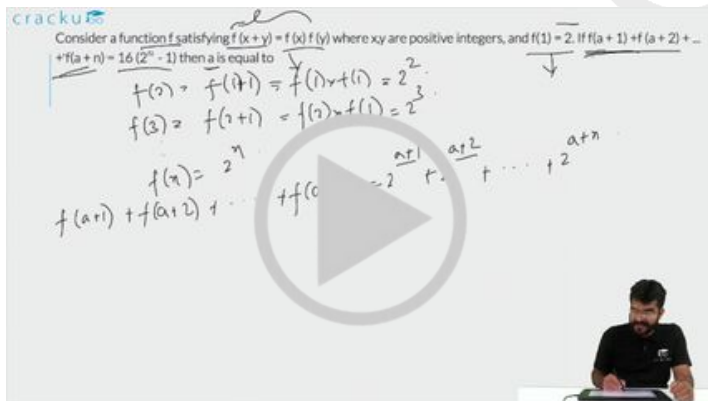
- A 0
- B 1
- C 2
- D 3



VIDEO SOLUTION

Question 47

Consider a function f satisfying $f(x+y) = f(x)f(y)$ where x, y are positive integers, and $f(1) = 2$. If $f(a+1) + f(a+2) + \dots + f(a+n) = 16(2^n - 1)$ then a is equal to



VIDEO SOLUTION

Question 48

Let $f(x) = x^2 + ax + b$ and $g(x) = f(x+1) - f(x-1)$. If $f(x) \geq 0$ for all real x , and $g(20) = 72$. then the smallest possible value of b is

- A 16

B 4

C 1

D 0

cracku

Let $f(x) = x^2 + ax + b$ and $g(x) = f(x+1) - f(x-1)$. If $f(x) \geq 0$ for all real x , and $g(2) = 72$, then the smallest possible value of b is

A) 16
C) 1

B) 4
D) 0

$\frac{2}{a} - 4b \leq 0$
 $b \geq \frac{2}{4}$
 $f(x+1) = (x+1)^2$

$\frac{ax^2 + bx + c}{b^2 - 4ac} \geq 0$



 VIDEO SOLUTION

100+ CAT DILR Questions

Question 49

The area of the region satisfying the inequalities $|x - y| \leq 1$, $y \geq 0$ and $y \leq 1$ is

cracku

The area of the region satisfying the inequalities $|x| - y \leq 1$, $y \geq 0$ and $y \leq 1$ is

$|x| - y \leq 1$
 $|x| \leq y + 1$
 $x > 0$
 $x \leq y + 1$
 $x = y + 1$

A hand-drawn graph on a Cartesian coordinate system. The x-axis and y-axis are shown. A horizontal line is drawn at y = 1. A horizontal line is also drawn at y = 0. Two lines, x = y + 1 and x = -y - 1, are drawn, forming a V-shape opening to the right. The region between y = 0 and y = 1, and between the two V-shaped lines, is shaded with diagonal lines. A large play button icon is overlaid on the graph.

 VIDEO SOLUTION

Question 50

In how many ways can a pair of integers (x, a) be chosen such that $x^2 - 2 \mid |x| + |a - 2| = 0$?

A 6

B 5

C 4

D 7

In how many ways can a pair of integers (x, a) be chosen such that $x^2 - 2|x| + |a - 2| = 0$?

Handwritten notes on the screen:

$$y^2 - 2y + p = 0$$

$$4 - 4p \geq 0$$

$$p \leq 1$$

$$|x| = y \geq 0$$

$$|a - 2| = p \geq 0$$

$$a^2 - 4a + c = 0$$

$$b^2 - 4ac \geq 0$$

A large play button icon is centered on the screen.

VIDEO SOLUTION

Question 51

In a group of 10 students, the mean of the lowest 9 scores is 42 while the mean of the highest 9 scores is 47. For the entire group of 10 students, the maximum possible mean exceeds the minimum possible mean by

- A 5
- B 4
- C 3
- D 6

cracku

In a group of 10 students, the mean of the lowest 9 scores is 42 while the mean of the highest 9 scores is 47. For the entire group of 10 students, the maximum possible mean exceeds the minimum possible mean by

Options: A) 5, B) 4, C) 3, D) 6

Handwritten notes on the screen:

$$42 \times 9 = 378$$

$$47 \times 9 = 423$$

$$10 - 1 = 9$$

$$10 = 45 + 1$$

$$378$$

$$423$$

A large play button icon is centered on the screen.

VIDEO SOLUTION

100+ CAT VARC Questions



Question 52

If $f(x + y) = f(x)f(y)$ and $f(5) = 4$, then $f(10) - f(-10)$ is equal to

- A 14.0625
- B 0
- C 15.9375

D 3

cracku

If $f(x+y) = f(x)f(y)$ and $f(5) = 4$, then $f(10) - f(-10)$ is equal to

A) 14.0625
B) 0
C) 15.9375
D) 3

$f(5) = 4$
 $f(10) = f(5+5) = f(5) \times f(5)$
 $= 4 \times 4 = 16$
 $f(-5) = 2K$
 $f(5) = f(-5+10) = f(-5) \times f(10)$

VIDEO SOLUTION

Question 53

If $f_1(x) = x^2 + 11x + n$ and $f_2(x) = x$, then the largest positive integer n for which the equation $f_1(x) = f_2(x)$ has two distinct real roots is

cracku

If $f_1(x) = x^2 + 11x + n$ and $f_2(x) = x$, then the largest positive integer n for which the equation $f_1(x) = f_2(x)$ has two distinct real roots is

$f_1(x) = f_2(x)$
 $x^2 + 11x + n = x$
 $x^2 + 10x + n = 0$

$ax^2 + bx + c = 0$
 $D > 0$

VIDEO SOLUTION

Question 54

Suppose $f(x, y)$ is a real-valued function such that $f(3x + 2y, 2x - 5y) = 19x$, for all real numbers x and y . The value of x for which $f(x, 2x) = 27$, is

cracku

Suppose $f(x, y)$ is a real-valued function such that $f(3x + 2y, 2x - 5y) = 19x$, for all real numbers x and y . The value of x for which $f(x, 2x) = 27$, is

Answer: _____

$3x + 2y = 2x - 5y$
 $6x + 4y = 2x - 5y$
 $4x = -9y$
 $x = -\frac{9}{4}y$

$f(3x + 2y, 2x - 5y) = 19x$
 $f(x, 2x) = 27$

VIDEO SOLUTION

CAT Expected Questions PDF

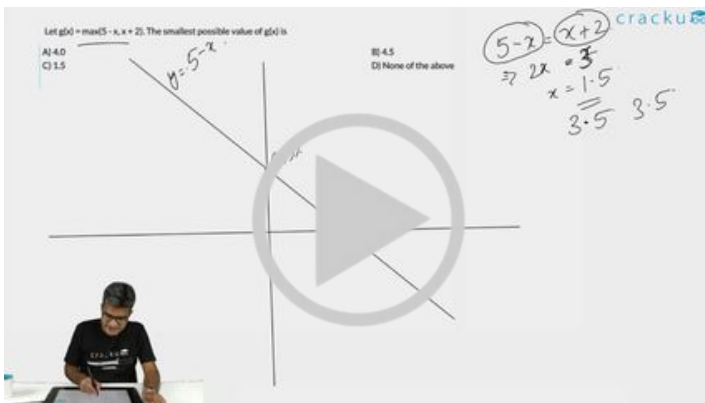
cracku



Question 55

Let $g(x) = \max(5 - x, x + 2)$. The smallest possible value of $g(x)$ is

- A 4.0
- B 4.5
- C 1.5
- D None of the above

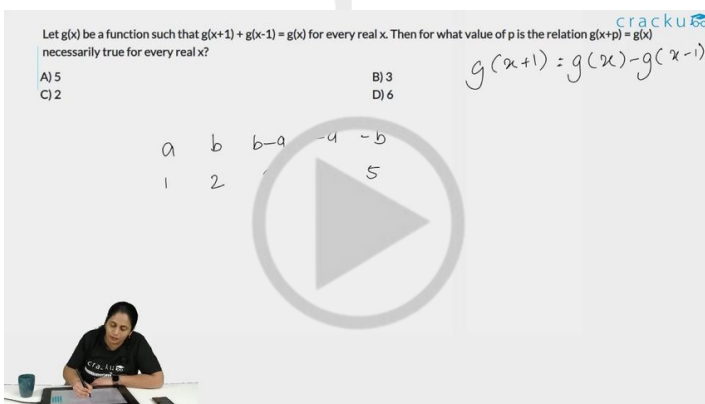


VIDEO SOLUTION

Question 56

Let $g(x)$ be a function such that $g(x+1) + g(x-1) = g(x)$ for every real x . Then for what value of p is the relation $g(x+p) = g(x)$ necessarily true for every real x ?

- A 5
- B 3
- C 2
- D 6

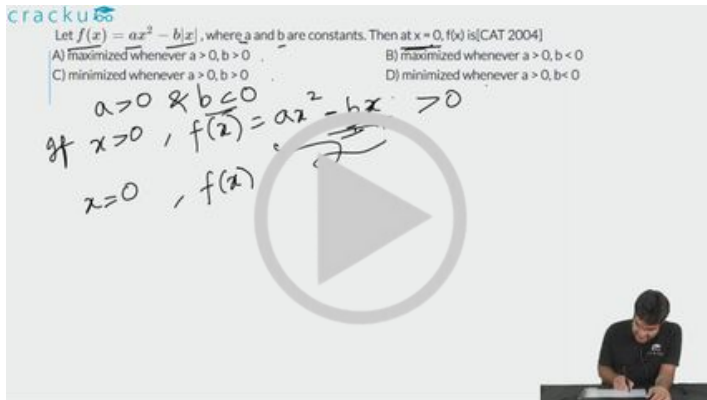


VIDEO SOLUTION

Question 57

Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then at $x = 0$, $f(x)$ is

- A maximized whenever $a > 0, b > 0$
- B maximized whenever $a > 0, b < 0$
- C minimized whenever $a > 0, b > 0$
- D minimized whenever $a > 0, b < 0$

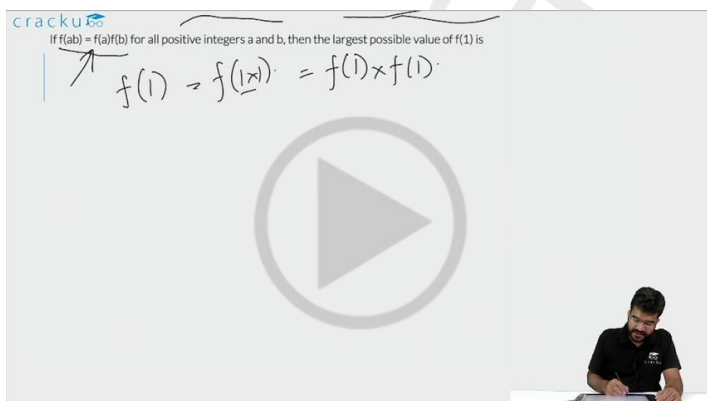


VIDEO SOLUTION



Question 58

If $f(ab) = f(a)f(b)$ for all positive integers a and b , then the largest possible value of $f(1)$ is



VIDEO SOLUTION

Question 59

The function $f(x) = |x - 2| + |2.5 - x| + |3.6 - x|$, where x is a real number, attains a minimum at

- A $x = 2.3$
- B $x = 2.5$
- C $x = 2.7$
- D None of the above

The function $f(x) = |x-2| + |2.5-x| + |3.6-x|$, where x is a real number, attains a minimum at

A) $x = 2.3$ B) $x = 2.5$
 C) $x = 2.7$ D) None of the above

Handwritten solution:

$f(x) = |x-2| + |x-2.5| + |x-3.6|$

$x < 2$ $2 \leq x \leq 2.5$ $2.5 < x < 3.6$ $x \geq 3.6$

$f(x) = x-2 + 2.5-x + 3.6-x = 4.1-3x$ (for $x < 2$)
 $f(x) = x-2 + x-2.5 + 3.6-x = 0.1$ (for $2 \leq x \leq 2.5$)
 $f(x) = x-2 + x-2.5 + 3.6-x = 0.1$ (for $2.5 < x < 3.6$)
 $f(x) = x-2 + x-2.5 + x-3.6 = 3x-8.1$ (for $x \geq 3.6$)

The minimum value is 0.1, which occurs for $2 \leq x \leq 2.5$. Therefore, the minimum is attained at $x = 2.5$.

VIDEO SOLUTION

Question 60

Suppose for any real number x , $[x]$ denotes the greatest integer less than or equal to x . Let $L(x, y) = [x] + [y] + [x + y]$ and $R(x, y) = [2x] + [2y]$. Then it is impossible to find any two positive real numbers x and y for which

- A $L(x, y) = R(x, y)$
- B $L(x, y) \neq R(x, y)$
- C $L(x, y) < R(x, y)$
- D $L(x, y) > R(x, y)$

VIEW SOLUTION

CAT WhatsApp Group



Question 61

If $a_1 = 1$ and $a_{n+1} - 3a_n + 2 = 4n$ for every positive integer n , then a_{100} equals

- A $3^{99} - 200$
- B $3^{99} + 200$
- C $3^{100} - 200$
- D $3^{100} + 200$

VIEW SOLUTION

Question 62

In the X-Y plane, the area of the region bounded by the graph of $|x+y| + |x-y| = 4$ is

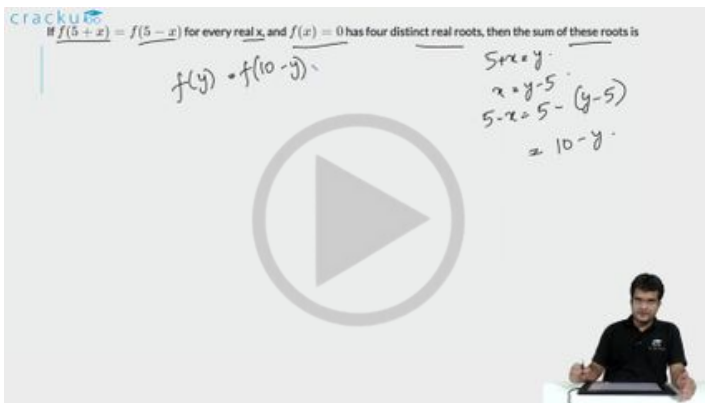
- A 8
- B 12
- C 16
- D 20

[VIEW SOLUTION](#)

Question 63

If $f(5+x) = f(5-x)$ for every real x , and $f(x) = 0$ has four distinct real roots, then the sum of these roots is

- A 0
- B 40
- C 10
- D 20



[VIDEO SOLUTION](#)



Question 64

Let $f(x)$ be a quadratic polynomial in x such that $f(x) \geq 0$ for all real numbers x . If $f(2) = 0$ and $f(4) = 6$, then $f(-2)$ is equal to

- A 12
- B 24
- C 6
- D 36

Let $f(x)$ be a quadratic polynomial in x such that $f(x) \geq 0$ for all real numbers x . If $f(2) = 0$ and $f(4) = 6$, then $f(-2)$ is equal to

A) 12
B) 24
C) 6
D) 36

$f(x) = a(x-b)^2 + c$
 $f(2) = 0$
 $f(x) = a(x-2)^2 + 0$
 $a > 0$

VIDEO SOLUTION

Question 65

A function f maps the set of natural numbers to whole numbers, such that $f(xy) = f(x)f(y) + f(x) + f(y)$ for all x, y and $f(p) = 1$ for every prime number p . Then, the value of $f(160000)$ is

- A 4095
- B 8191
- C 2047
- D 1023

A function f maps the set of natural numbers to whole numbers, such that $f(xy) = f(x)f(y) + f(x) + f(y)$ for all x, y and $f(p) = 1$ for every prime number p . Then, the value of $f(160000)$ is

A) 4095
B) 8191
C) 2047
D) 1023

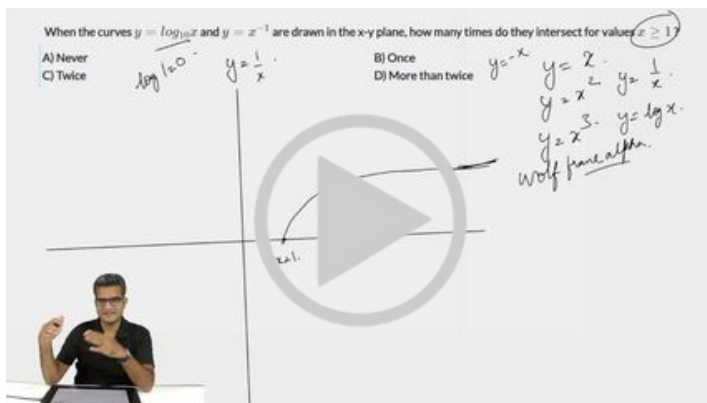
$f(p) = 1$
 $f(p^2) = 1 \times 1 + 1 + 1 = 3$
 $160000 = 16 \times 10000$
 $= 2^4 \times 100^2$
 $= 2^4 \times 10^4$
 $= 2^8 \times 5^4$

VIDEO SOLUTION

Question 66

When the curves $y = \log_{10}x$ and $y = x^{-1}$ are drawn in the x - y plane, how many times do they intersect for values $x \geq 1$?

- A Never
- B Once
- C Twice
- D More than twice



VIDEO SOLUTION



Instructions

Directions for the next 2 questions:

For real numbers x, y , let

$f(x, y)$ = Positive square-root of $(x + y)$, if $(x + y)^{0.5}$ is real

$f(x, y) = (x + y)^2$; otherwise

$g(x, y) = (x + y)^2$, if $\sqrt{(x + y)}$ is real

$g(x, y) = -(x + y)$ otherwise

Question 67

Under which of the following conditions is $f(x, y)$ necessarily greater than $g(x, y)$?

- A Both x and y are less than -1
- B Both x and y are positive
- C Both x and y are negative
- D $y > x$

VIEW SOLUTION

Question 68

Which of the following expressions yields a positive value for every pair of non-zero real numbers (x, y) ?

- A $f(x, y) - g(x, y)$
- B $f(x, y) - (g(x, y))^2$
- C $g(x, y) - (f(x, y))^2$
- D $f(x, y) + g(x, y)$

VIEW SOLUTION

Instructions

Directions for the following two questions:

Let S be the set of all pairs (i, j) where $1 \leq i < j \leq n$, and $n \geq 4$ (i and j are natural numbers). Any two distinct members of S are called "friends" if they have one constituent of the pairs in common and "enemies" otherwise.

For example, if $n = 4$, then $S = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$. Here, $(1, 2)$ and $(1, 3)$ are friends, $(1, 2)$ and $(2, 3)$ are also friends, but $(1, 4)$ and $(2, 3)$ are enemies.

Question 69

For general n , consider any two members of S that are friends. How many other members of S will be common friends of both these members?

- A $\frac{(n^2 - 5n + 8)}{2}$
- B $2n - 6$
- C $\frac{n(n-3)}{2}$
- D $n - 2$
- E $\frac{(n^2 - 7n + 16)}{2}$

 [VIEW SOLUTION](#)



CAT Previous Papers

with detailed video solutions and analysis



Question 70

For general n , how many enemies will each member of S have?

- A $n - 3$
- B $(1/2)(n^2 - 3n - 2)$
- C $2n - 7$
- D $(1/2)(n^2 - 5n + 6)$
- E $(1/2)(n^2 - 7n - 14)$

 [VIEW SOLUTION](#)

Instructions

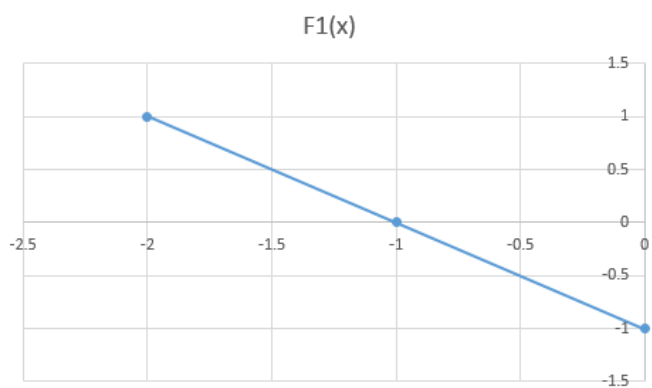
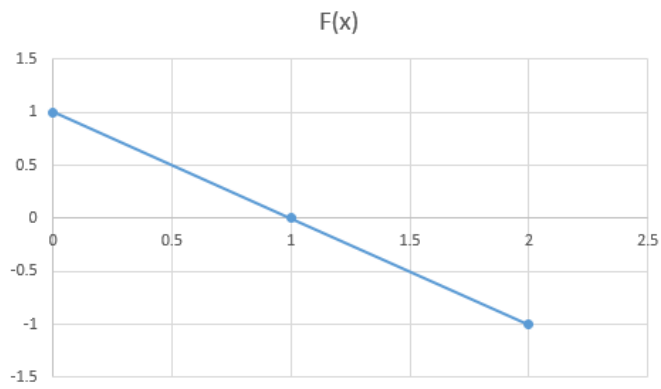
DIRECTIONS for the following questions: These questions are based on the situation given below: In each of the questions a pair of graphs $F(x)$ and $F1(x)$ is given. These are composed of straight-line segments, shown as solid lines, in the domain $x \in (-2, 2)$.

choose the answer as

- a. If $F1(x) = -F(x)$
- b. if $F1(x) = F(-x)$
- c. if $F1(x) = -F(-x)$

d. if none of the above is true

Question 71



A a

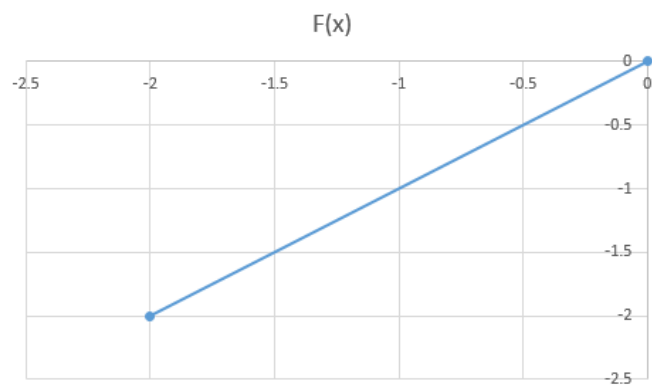
B b

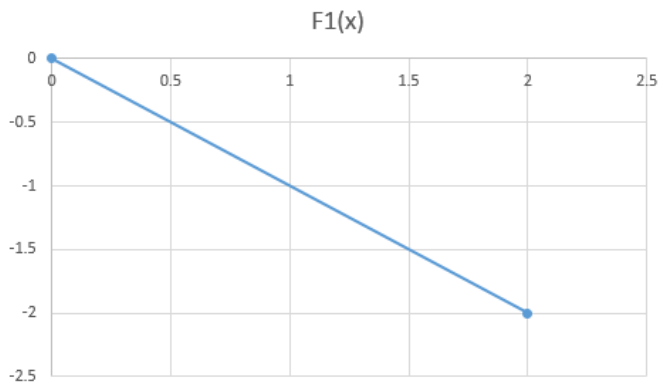
C c

D d

[VIEW SOLUTION](#)

Question 72





- A a
- B b
- C c
- D d

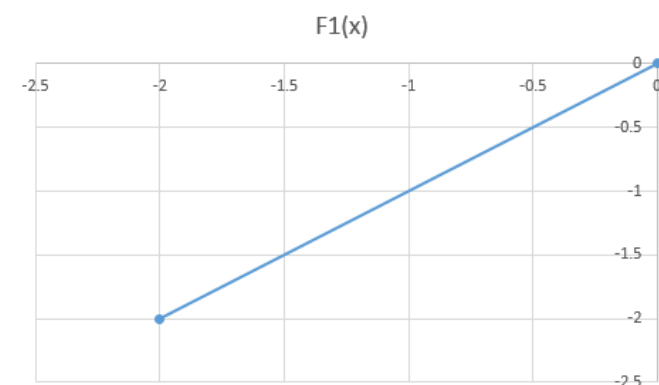
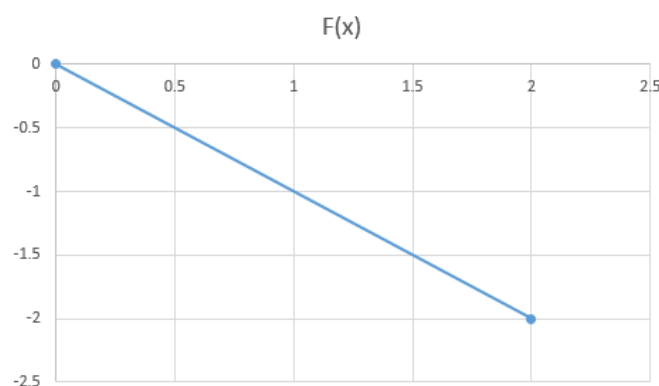
[VIEW SOLUTION](#)



CAT Syllabus PDF



Question 73

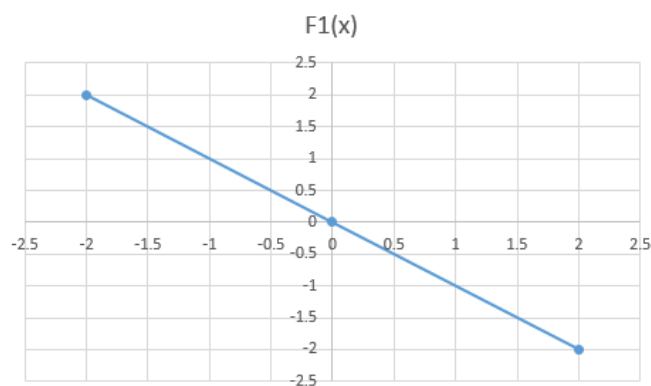
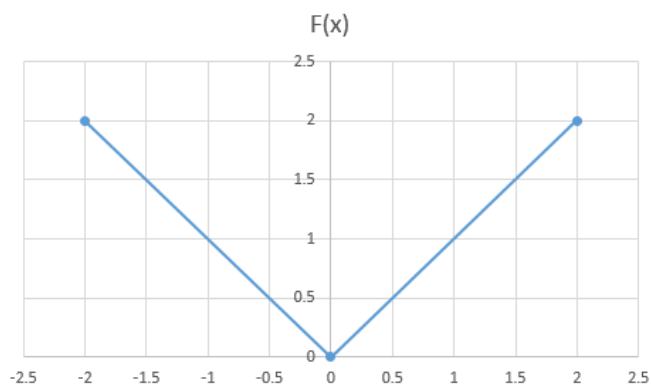


- A a
- B b
- C c

D d

[VIEW SOLUTION](#)

Question 74



A a

B b

C c

D d

[VIEW SOLUTION](#)

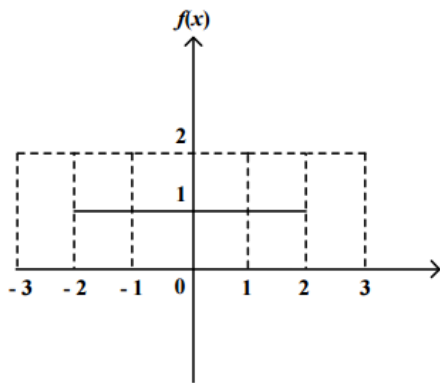
Instructions

Directions for the next 3 questions:

Given below are three graphs made up of straight-line segments shown as thick lines. In each case choose the answer as:

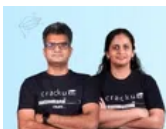
- a) if $f(x) = 3f(-x)$
- b) if $f(x) = -f(-x)$
- c) if $f(x) = f(-x)$
- d) if $3f(x) = 6f(-x)$, for $x \geq 0$

Question 75



- A if $f(x) = 3f(-x)$
- B if $f(x) = -f(-x)$
- C if $f(x) = f(-x)$
- D if $3f(x) = 6f(-x)$, for $x \geq 0$

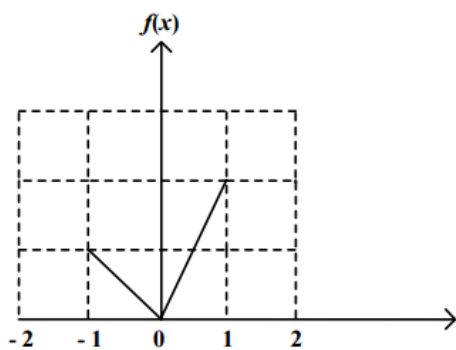
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CAT Formula PDF



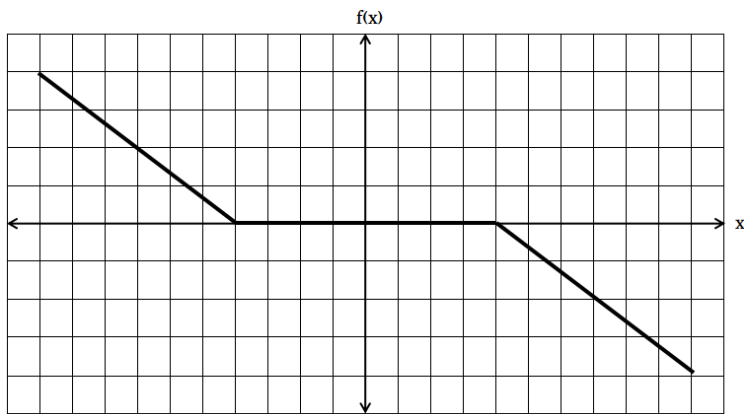
Question 76



- A if $f(x) = 3f(-x)$
- B if $f(x) = -f(-x)$
- C if $f(x) = f(-x)$
- D if $3f(x) = 6f(-x)$, for $x \geq 0$

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Question 77



- A if $f(x) = 3f(-x)$
- B if $f(x) = -f(-x)$
- C if $f(x) = f(-x)$
- D if $3f(x) = 6f(-x)$, for $x \geq 0$

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Instructions

Directions for the next 2 questions:

For a real number x , let

$$f(x) = 1/(1+x), \text{ if } x \text{ is non-negative}$$

$$f(x) = 1+x, \text{ if } x \text{ is negative}$$

$$f^n(x) = f(f^{n-1}(x)), n = 2, 3, \dots$$

Question 78

What is the value of the product, $f(2)f^2(2)f^3(2)f^4(2)f^5(2)$?

- A $1/3$
- B 3
- C $1/18$
- D None of these

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Question 79

r is an integer ≥ 2 . Then, what is the value of $f^{r-1}(-r) + f^r(-r) + f^{r+1}(-r)$?

- A -1
- B 0

C 1

D None of these

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Instructions

DIRECTIONS for the following questions:

These questions are based on the situation given below: A robot moves on a graph sheet with x and y-axes. The robot is moved by feeding it with a sequence of instructions. The different instructions that can be used in moving it, and their meanings are: Instruction Meaning GOTO(x,y) move to point with coordinates (x, y) no matter where you are currently WALKX(P) Move parallel to the x-axis through a distance of p, in the positive direction if p is positive, and in the negative direction if p is negative WALKY(P) Move parallel to the y-axis through a distance of p, in the positive direction if p is positive, and in the negative direction if p is negative.

Question 80

The robot is initially at (x, y), $x > 0$ and $y < 0$. The minimum number of instructions needed to be executed to bring it to the origin (0,0) if you are prohibited from using the GOTO instruction is:

A 2

B 1

C $x + y$

D 0

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Question 81

The robot reaches point (6, 6) when a sequence of three instructions is executed, the first of which is a GOTO(x, y) instruction, the second is WALKX(2) and the third is WALKY(4). What are the values of x and y?

A 2, 4

B 0, 0

C 4, 2

D 2, 2

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CAT Percentile Predictor



Instructions

Answer the questions based on the following information. A, S, M and D are functions of x and y, and they are defined as follows.

$$A(x, y) = x + y$$

$$S(x, y) = x - y$$

$$M(x, y) = xy$$

$$D(x, y) = \frac{x}{y}, y \neq 0$$

Question 82

What is the value of $S[M(D(A(a, b), 2), D(A(a, b), 2)), M(D(S(a, b), 2), D(S(a, b), 2))]$?

- A $a^2 + b^2$
- B ab
- C $a^2 - b^2$
- D $\frac{a}{b}$

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Question 83

What is the value of $M(M(A(M(x, y), S(y, x)), x), A(y, x))$ for $x = 2, y = 3$?

- A 60
- B 140
- C 25
- D 70

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Instructions

Directions for the next 3 questions: For three distinct real positive numbers x, y and z , let

$$f(x, y, z) = \min(\max(x, y), \max(y, z), \max(z, x))$$

$$g(x, y, z) = \max(\min(x, y), \min(y, z), \min(z, x))$$

$$h(x, y, z) = \max(\max(x, y), \max(y, z), \max(z, x))$$

$$j(x, y, z) = \min(\min(x, y), \min(y, z), \min(z, x))$$

$$m(x, y, z) = \max(x, y, z)$$

$$n(x, y, z) = \min(x, y, z)$$

Question 84

Which of the following is necessarily greater than 1?

- A $(h(x, y, z) - f(x, y, z)) / j(x, y, z)$
- B $j(x, y, z) / h(x, y, z)$
- C $f(x, y, z) / g(x, y, z)$
- D $(f(x, y, z) + h(x, y, z) - g(x, y, z)) / j(x, y, z)$

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CAT Score Calculator



Question 85

Which of the following expressions is indeterminate?

- A $(f(x, y, z) - h(x, y, z)) / (g(x, y, z) - j(x, y, z))$
- B $[f(x, y, z) + h(x, y, z) + g(x, y, z) + j(x, y, z)] / [j(x, y, z) + h(x, y, z) - m(x, y, z) - n(x, y, z)]$
- C $[g(x, y, z) - j(x, y, z)] / [f(x, y, z) - h(x, y, z)]$
- D $[h(x, y, z) - f(x, y, z)] / [n(x, y, z) - g(x, y, z)]$

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Question 86

Which of the following expressions is necessarily equal to 1?

- A $(f(x, y, z) - m(x, y, z)) / (g(x, y, z) - h(x, y, z))$
- B $(m(x, y, z) - f(x, y, z)) / (g(x, y, z) - n(x, y, z))$
- C $(j(x, y, z) - g(x, y, z)) / h(x, y, z)$
- D $(f(x, y, z) - h(x, y, z)) / f(x, y, z)$

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Instructions

DIRECTIONS for the following questions: These questions are based on the situation given below: Let x and y be real numbers

$$f(x, y) = |x + y|$$

$$F(f(x, y)) = -f(x, y)$$

$$G(f(x, y)) = -F(f(x, y))$$

Question 87

Which of the following statements is true?

- A $F(f(x, y)) \cdot G(f(x, y)) = -F(f(x, y)) \cdot G(f(x, y))$
- B $F(f(x, y)) \cdot G(f(x, y)) > -F(f(x, y)) \cdot G(f(x, y))$
- C $F(f(x, y)) \cdot G(f(x, y)) \neq G(f(x, y)) \cdot F(f(x, y))$
- D $F(f(x, y)) + G(f(x, y)) + f(x, y) = f(-x, -y)$

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Question 88

What is the value of $f(G(f(1, 0)), f(F(f(1, 2)), G(f(1, 2))))$?

- A 3
- B 2
- C 1
- D 0

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Question 89

Which of the following expressions yields x^2 as its result?

- A $F(f(x, -x)).G(f(x, -x))$
- B $F(f(x, x)).G(f(x, x)).4$
- C $-F(f(x, x).G(f(x, x))) / \log_2 16$
- D $f(x, x).f(x, x)$

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Instructions

Directions for the following two questions:

Answer the questions on the basis of the information given below.

$$f_1(x) = x \text{ if } 0 \leq x \leq 1 \quad f_1(x) = 1 \text{ if } x > 1 \quad f_1(x) = 0 \text{ otherwise}$$

$$f_2(x) = f_1(-x) \text{ for all } x$$

$$f_3(x) = -f_2(x) \text{ for all } x$$

$$f_4(x) = f_3(-x) \text{ for all } x$$

Question 90

Which of the following is necessarily true?

- A $f_4(x) = f_1(x)$ for all x
- B $f_1(x) = -f_3(-x)$ for all x
- C $f_2(-x) = f_4(x)$ for all x
- D $f_1(x) + f_3(x) = 0$ for all x

Directions for the following two questions:
Answer the questions on the basis of the information given below.

$f_1(x) = x$ if $0 \leq x \leq 1$
 $f_1(x) = 1$ if $x > 1$
 $f_1(x) = 0$ otherwise

$f_2(x) = f_1(-x)$ for all x
 $f_3(x) = -f_2(x)$ for all x
 $f_4(x) = f_3(-x)$ for all x

How many of the following products are necessarily zero for every x :

$f_1(x)f_2(x), f_2(x)f_3(x), f_3(x)f_4(x)$

A) 0
B) 1
C) 2

Handwritten notes on the right:

$f_1(x)$
 $f_2(x) = f_1(-x)$
 $f_3(x) = -f_1(-x)$
 $f_4(x) = -f_1(x)$

Handwritten calculations for the products:

① $f_1(x)f_2(x) = f_1(x)f_1(-x)$
 ② $f_2(x)f_3(x) = f_1(-x)f_1(x) = f_1(x)f_1(-x)$
 ③ $f_3(x)f_4(x) = -f_1(-x)f_1(x) = -f_1(x)f_1(-x)$

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Question 91

How many of the following products are necessarily zero for every x :

$f_1(x)f_2(x), f_2(x)f_3(x), f_2(x)f_4(x)$

- A) 0
B) 1
C) 2
D) 3

Directions for the following two questions:
Answer the questions on the basis of the information given below.

$f_1(x) = x$ if $0 \leq x \leq 1$
 $f_1(x) = 1$ if $x > 1$
 $f_1(x) = 0$ otherwise

$f_2(x) = f_1(-x)$ for all x
 $f_3(x) = -f_2(x)$ for all x
 $f_4(x) = f_3(-x)$ for all x

How many of the following products are necessarily zero for every x :

$f_1(x)f_2(x), f_2(x)f_3(x), f_2(x)f_4(x)$

A) 0
B) 1
C) 2

Handwritten notes on the right:

$f_1(x)$
 $f_2(x) = f_1(-x)$
 $f_3(x) = -f_1(-x)$
 $f_4(x) = -f_1(x)$

Handwritten calculations for the products:

① $f_1(x)f_2(x) = f_1(x)f_1(-x)$
 ② $f_2(x)f_3(x) = f_1(-x)f_1(x) = f_1(x)f_1(-x)$
 ③ $f_2(x)f_4(x) = f_1(-x)f_1(x) = f_1(x)f_1(-x)$

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Instructions

Directions for the following two questions:

Let $a_1 = p$ and $b_1 = q$, where p and q are positive quantities.

Define $a_n = pb_{n-1}$, $b_n = qb_{n-1}$, for even $n > 1$. and $a_n = pa_{n-1}$, $b_n = qa_{n-1}$, for odd $n > 1$.

Question 92

Which of the following best describes $a_n + b_n$ for even n ?

A $q(pq)^{(n/2)-1}(p+q)^{n/2}$

B $q(pq)^{(n/2)-1}(p+q)$

C $qp^{n/2}(p+q)$

D $q^{n/2}(p+q)$

E $q^{n/2}(p+q)^{n/2}$

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Question 93

If $p = 1/3$ and $q = 2/3$, then what is the smallest odd n such that $a_n + b_n < 0.01$?

A 7

B 13

C 11

D 9

E 15

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Answers

1. 12	2. D	3. 2	4. A	5. 32	6. A	7. E	8. B
9. B	10. D	11. 2	12. D	13. C	14. C	15. C	16. B
17. 54	18. B	19. B	20. D	21. D	22. 45	23. B	24. E
25. D	26. C	27. A	28. A	29. B	30. D	31. B	32. 10
33. C	34. 44	35. 20	36. C	37. D	38. A	39. 12	40. B
41. B	42. B	43. E	44. D	45. D	46. C	47. 3	48. B
49. 3	50. D	51. B	52. C	53. 24	54. 3	55. D	56. D
57. D	58. 1	59. B	60. D	61. C	62. C	63. D	64. B
65. A	66. B	67. A	68. D	69. D	70. D	71. C	72. B
73. B	74. D	75. C	76. D	77. B	78. C	79. B	80. A
81. C	82. B	83. D	84. D	85. B	86. A	87. D	88. C
89. C	90. B	91. C	92. B	93. D			

Explanations

1. **12**

Given,

$$f(x) + f(x-1) = 1 \text{ (1)}$$

$$f(x^2 - x) = 5 \text{ (2)}$$

$$g(x) = x^2$$

Substituting $x = 1$ in (1) and (2), we get

$$f(0) = 5$$

$$f(1) + f(0) = 1$$

$$f(1) = 1 - 5 = -4$$

$$f(2) + f(1) = 1$$

$$f(2) = 1 + 4 = 5$$

$$f(n) = 5 \text{ if } n \text{ is even and } f(n) = -4 \text{ if } n \text{ is odd}$$

$$f(g(5)) + g(f(5)) = f(25) + g(-4) = -4 + 16 = 12$$

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2. **D**

Set $A = \{2, 3, 5, 7, 11, 13\}$ so $|A| = 6$

Set $B = \{1, 8, 27\}$ so $|B| = 3$

Without any restrictions, each element in A can map to any of the 3 elements in B. Thus, the total number of functions is: $3^6 = 729$

Excluding Functions That Miss One Element in B: If a function does not map to an element in B, there are 2 elements in B left for mapping. The total number of such functions (for each specific element not mapped) is: $2^6 = 64$

Since there are 3 elements in B, the total number of such functions is: $3 \times 64 = 192$

Adding Back Functions That Miss Two Elements in B: If a function misses two elements in B, there is only 1 element left for mapping. The total number of such functions is: $1^6 = 1$.

Since there are 3C_2 ways to choose which two elements are missed, the total number of such functions is: 3

Using the inclusion-exclusion principle, the number of functions where all elements of B are mapped by at least one element of A is:

$$729 - 192 + 3 = 540.$$

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3.2

The expression on the right-hand side will have two critical points: 2 and -2

For any value of **x greater than equal to 2**, the equation changes to $x+2-(x-2) = 4$

the value of $\min\{x, 2\}$ would be 2, so we would want $\max\{x, 2\}$ to be $4+2 = 6$

Therefore, $x=6$ works.

For any value of **x less than equal to -2**, the equation changes to $-(x+2)+(x-2) = -4$

the value of $\max\{x, 2\}$ would be 2, so we would want $\min\{x, 2\}$ to be 6 again; this cannot be the case.

In general, subtracting the smaller (min) of the two values from the bigger (max) can not lead to a negative number. The max it can lead to is a 0

When **x lies between -2 and 2**, the equation becomes $2x$

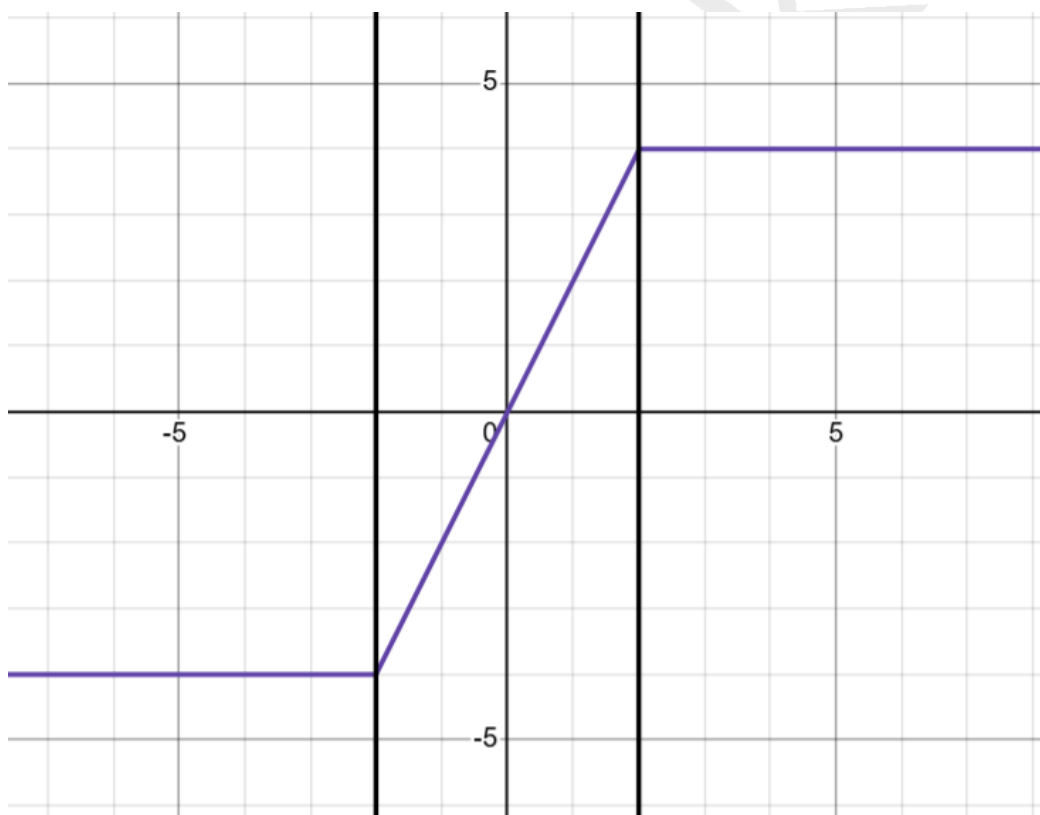
The maximum function will give back 2, and the minimum function will give back x, with the right-hand side giving $2x$

Solving this we would get $2-x = 2x$ which is $x=2/3$

Therefore, there are two real values of x for which the given equation holds.

Hence, 2 is the correct answer.

The graph of the function on the right hand side can be visualised as:



Hence, 2 is the correct answer.

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4.A

Consider the first term:

$$\sqrt{1 + 1/1^2 + 1/2^2} = \sqrt{9/4} = 3/2$$

$$\text{Second term: } \sqrt{1 + 1/2^2 + 1/3^2} = \sqrt{49/36} = 7/6$$

$$\text{First term} + \text{Second term} = 3/2 + 7/6 = (9+7)/6 = 16/6 = 8/3 = 3 - 1/3$$

.

.

$$\text{Required sum} = 2008 - 1/2008$$

 VIDEO SOLUTION

5.32

$$f(x) = \min(2x^2, 52 - 5x)$$

The maximum possible value of this function will be attained at the point in which $2x^2$ is equal to $52 - 5x$.

$$2x^2 = 52 - 5x$$

$$2x^2 + 5x - 52 = 0$$

$$(2x + 13)(x - 4) = 0$$

$$\Rightarrow x = \frac{-13}{2} \text{ or } x = 4$$

It has been given that x is a positive real number. Therefore, we can eliminate the case $x = \frac{-13}{2}$.

$x = 4$ is the point at which the function attains the maximum value. 4 is not the maximum value of the function.

Substituting $x = 4$ in the original function, we get, $2x^2 = 2 * 4^2 = 32$.

$$f(x) = 32.$$

Therefore, 32 is the right answer.

 VIDEO SOLUTION

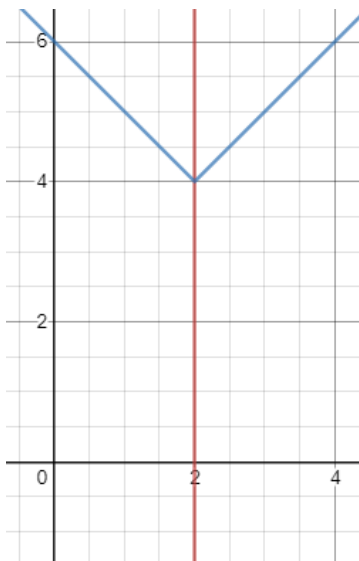




CAT Sectional Tests



6.A



The required figure is a trapezium with vertices A(0,0), B(2,0), C(2,4) and D(0,6)

$$AB = 2 \quad BC = 4 \quad \text{and} \quad AD = 6$$

$$\text{Area of trapezium} = \frac{1}{2} (\text{sum of the opposite sides}) \cdot \text{height} = \frac{1}{2} (4 + 6) \cdot 2 = 10$$

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7.E

The minimum value is obtained when $2x+1 = 3-4x \Rightarrow 6x = 2 \Rightarrow x = 1/3$

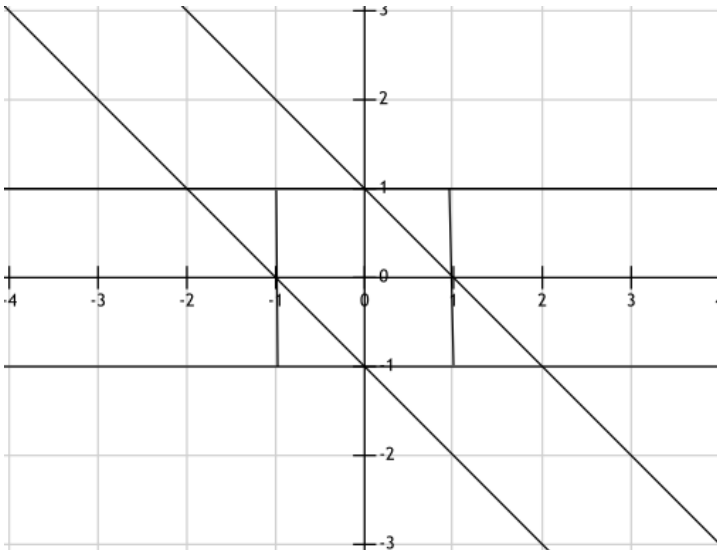
So, $f(x) = 2 \cdot 1/3 + 1 = 5/3$

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8.B

$|x| = 1$ and $|y| = 1$ form a square of area $= 2 \cdot 2 = 4$ sq units

$|x+y| = 1$ forms a set of parallel lines cutting the axes at $(1,0)$, $(0,1)$, $(-1,0)$ and $(0,-1)$. The graph is as shown:



The area bounded by the three curves is $2 \cdot 2 - 1/2 \cdot 1 \cdot 1 \cdot 2 = 4 - 1 = 3$ sq units

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9.B

For $f(1,2)$. First consider $x=0$ and $y=1$ and use 3rd given equation, we get $f(0, f(1,1))$ now for $f(1,1)$ take $x=0$ and $y=0$ we get $f(0, f(1,0))$, for $f(1,0)$ which we use 2nd equation we get $f(0,1)$ whose value is 2. So we have $f(0, f(1,0)) = f(0,2)$ whose value is 3 then put this in $f(0, f(1,1))$ we get $f(0,3)$ we get as 4

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10.D

For $x < 0$; $y = x + 2$

for $0 < x < \frac{1}{2}$; $y = x + 2$

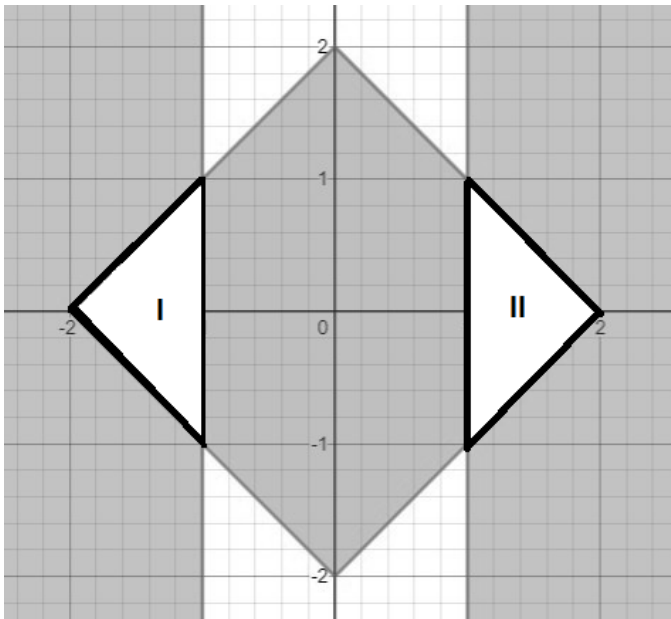
for $x > \frac{1}{2}$; $y = 3 - x$

Hence, y attains its maxima at $x = \frac{1}{2}$ i.e. $y = 2.5$

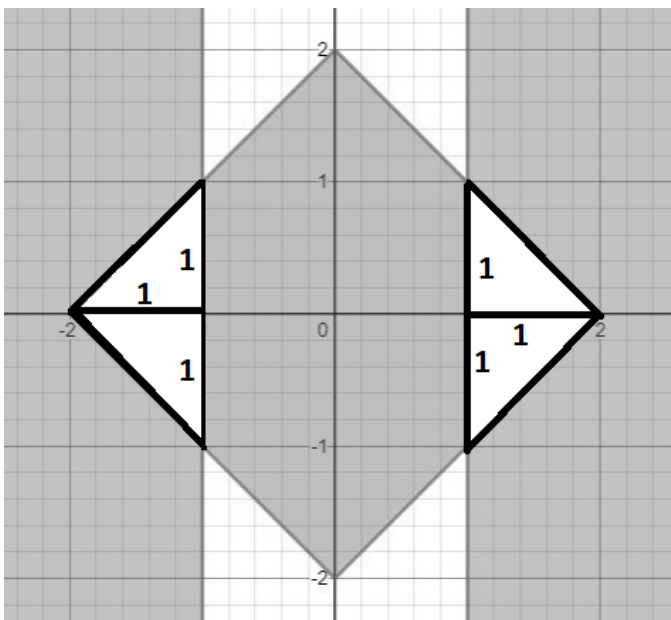
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11.2



Sum of the area of region I and II is the required area.



Now, required area = $4 \times \frac{1}{2} \times 1 \times 1 = 2$

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12. D

Among all options only D satisfies the given equations as follows:

$$f(y) = \frac{1-y}{1+y}$$

and for x:

$$y + xy = 1 - x$$

$$x(1 + y) = 1 - y$$

$$x = \frac{1-y}{1+y}$$

Hence $x = f(y)$

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13. C

On solving both equations, we will get $x = \frac{23}{2+2k}$
 now for having no solutions to system $2+2k$ should be 0.
 Hence $k=-1$

 VIDEO SOLUTION

14. C

We are given, $f(x) + 2f\left(\frac{1}{x}\right) = 3x$

Substituting $\frac{1}{x}$ for x

$$f\left(\frac{1}{x}\right) + 2f(x) = \frac{3}{x}$$

Multiplying the second equation by 2 we will have

$$2f\left(\frac{1}{x}\right) + 4f(x) = \frac{6}{x}$$

Subtracting the first equation from the second equation we have,

$$3f(x) = \frac{6}{x} - 3x$$

$$f(x) = \frac{2}{x} - x$$

We want the sum of values when this function equals 3,

$$\frac{2}{x} - x = 3$$

$$x^2 + 3x - 2 = 0$$

Since the discriminant is greater than zero, both values of x will be real, and we can directly take the sum of values of x to be,

$$-\frac{3}{1}$$

Answer is -3.

 VIDEO SOLUTION

15. C

$$a = r(b+c)$$

$$b = r(a+c)$$

$$c = r(a+b)$$

On adding all the equations,

$$a+b+c = 2r(a+b+c)$$

$$\text{If } r = 1/2, a+b+c = a+b+c \text{ (valid)}$$

$$\text{If } r = -1, a+b+c = -2(a+b+c) \Rightarrow a+b+c = 0 \Rightarrow b+c = -a \text{ and } a/(b+c) = a/(-a) = -1 \text{ (valid)}$$

So, r can take the values $1/2$ or -1

 VIDEO SOLUTION



16. B

Putting y as x and x as y in given options, only options B satisfies the function property as follows.

$$x = \frac{2y+3}{3y-2}$$

$$\text{or } 3xy - 2x = 2y+3$$

$$\text{or } y(3x-2) = 2x+3$$

$$\text{or } y = \frac{2x+3}{3x-2}$$

 VIEW SOLUTION

17. 54

$$f(x+2) = f(x) + f(x+1)$$

As we can see, the value of a term is the sum of the 2 terms preceding it.

It has been given that $f(11) = 91$ and $f(15) = 617$.

We have to find the value of $f(10)$.

$$\text{Let } f(10) = b$$

$$f(12) = b + 91$$

$$f(13) = 91 + b + 91 = 182 + b$$

$$f(14) = 182 + b + 91 + b = 273 + 2b$$

$$f(15) = 273 + 2b + 182 + b = 455 + 3b$$

It has been given that $455 + 3b = 617$

$$3b = 162$$

$$\Rightarrow b = 54$$

Therefore, 54 is the correct answer.

 VIDEO SOLUTION

18. B

Let p and q be any two elements of the set A.

For the computation of the GCF of elements of the set A, we can replace both p and q by just the GCF(p,q) and the result is unchanged.

So, for every application of the function h, we are reducing the number of elements of the set A by 1. (In this case two numbers p and q are replaced by one number GCF(p,q)).

Expanding this concept further, the minimum number of times the function h should be called is n-1

 VIEW SOLUTION

19. B

$y = a + bx$ is a linear function and clearly the function shown in the table is not linear.

Lets take option B:

$$\text{Suppose } y = f(x) = a + bx + cx^2$$

$$f(1)=4 \Rightarrow a+b+c=4 \text{ -- (1)}$$

$$f(2)=8 \Rightarrow a+2b+4c=8 \text{ -- (2)}$$

$$f(3)=14 \Rightarrow a+3b+9c=14 \text{ -- (3)}$$

Let us solve the equations and see if $f(4)$, $f(5)$, $f(6)$ also satisfy the given equation.

$$(2)-(1) \Rightarrow b+3c=4$$

$$(3)-(2) \Rightarrow b+5c=6$$

This gives $c=1$ and $b=1 \Rightarrow a=2$

$$\text{So, } f(x) = 2 + x + x^2$$

$$f(4) = 2 + 4 + 4^2 = 22$$

$$f(5) = 2 + 5 + 5^2 = 32$$

$$f(6) = 2 + 6 + 6^2 = 44$$

As all the three also satisfy the numbers given in the table, it can be inferred that the relationship between x and y is quadratic and the correct option is option (b)

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20. D

For a normal graph with y and x -axis, the equation of the line passing through the origin is $y=mx$ where m is the slope of the line.

m is +ve if the angle made by the line with the x -axis is $< 90^\circ$

$\therefore$ The equation of the line in the given graph would be $y-x = k(y+x)$ since the axes are $y-x$ and $y+x$ and the line is passing through the origin.

$k > 1$ because the angle is greater than 45°

$$y = \frac{x(k+1)}{1-k}$$

Since $k > 1$

Therefore $y < 0$ for $x > -1$ and $y > 0$ for $x < -1$

Option d correctly satisfy this condition

[VIEW SOLUTION](#)

100+ CAT Quant Questions



21. D

$$|f(x) + g(x)| = |f(x)| + |g(x)| \text{ if and only if}$$

$$\text{case 1: } f(x) \geq 0 \text{ and } g(x) \geq 0$$

$$\Leftrightarrow 2x - 5 \geq 0 \text{ and } 7 - 2x \geq 0$$

$$\Leftrightarrow x \geq \frac{5}{2} \text{ and } \frac{7}{2} \geq x$$

$$\Leftrightarrow \frac{5}{2} \leq x \leq \frac{7}{2}$$

$$\text{case 2: } f(x) \leq 0 \text{ and } g(x) \leq 0$$

$$\Leftrightarrow 2x - 5 \leq 0 \text{ and } 7 - 2x \leq 0$$

$$\Leftrightarrow x \leq \frac{5}{2} \text{ and } \frac{7}{2} \leq x$$

So $x \leq 5/2$ and $x \geq 7/2$ which is not possible.

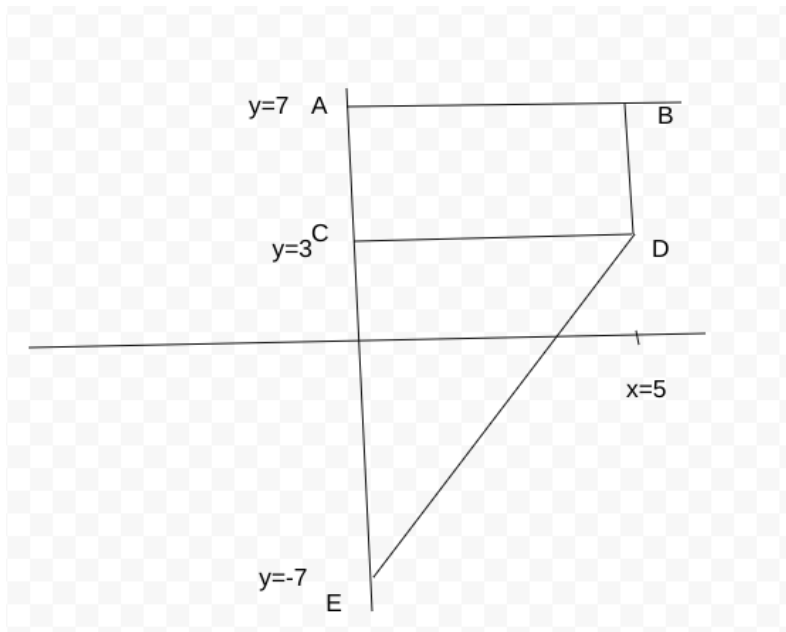
Hence, answer is

$$\Leftrightarrow \frac{5}{2} \leq x \leq \frac{7}{2}$$

VIDEO SOLUTION

22.45

From the inequality and nature of x , and y , we get the given diagram:



We need to find the area of the quadrilateral $ABDE$ = area of rectangle $ABCD$ + area of triangle CDE

$\Rightarrow$ Area of $ABCD$ = $(7-3)*5 = 20$ units, and the area of triangle CDE = $(1/2)*10*5 = 25$ units.

Hence, the area of the quadrilateral $ABDE$ = $(20+25) = 45$ units.

VIDEO SOLUTION

23. B

The union of the two sets is the set of positive integers. Also, given the increasing nature of elements, either $f(1)$ or $g(1)$ must be equal to 1. If $g(1) = 1$, then $f(f(1)) = 0$ which cannot be under the given conditions.

Hence, $f(1) = 1$

$g(1) = f(f(1))+1 = 2$

VIEW SOLUTION

24. E

For seed $(n) = 9$, all the numbers below 500 must have a digit sum of 9.

These numbers are all divisible by 9.

So total number of numbers below 500 and divisible by 9 is 55.

VIDEO SOLUTION

25. D

Let $(x^2 + 3x)$ be equal to k . We have,

$$f(x) = k(k+2) = k^2 + 2k$$

$$\text{Therefore, } \sqrt{f(x)+1} = \sqrt{k^2+2k+1} = \sqrt{(k+1)^2} = k+1 = 9701$$

We get $k = 9700$.

$$\text{Thus, } x^2 + 3x = 9700 \text{ or } x^2 + 3x - 9700 = 0$$

Since x is real, the discriminant of the above quadratic has to be greater than or equal to zero.

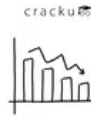
We find that $3^2 + 4 * 9700 \geq 0$ and therefore the quadratic has real roots.

The sum of the roots will be $-\frac{b}{a} = -\frac{3}{1} = -3$

Option D is the correct answer.

 [VIEW SOLUTION](#)

100+ CAT DILR Questions



26. C

$$\text{Let } y = \frac{2x - 3}{2x^2 + 4x - 6}$$

$$\Rightarrow 2yx^2 + 4yx - 6y = 2x - 3$$

$$\Rightarrow 2yx^2 + (4y - 2)x - 6y + 3 = 0$$

Equation (1) is a quadratic in x , where x is real. Therefore, the discriminant of the quadratic has to be greater than or equal to zero.

$$(4y - 2)^2 + 4 \times 2y \times (6y - 3) \geq 0$$

$$16y^2 + 4 - 16y - 24y + 48y^2 \geq 0$$

$$64y^2 - 40y + 4 \geq 0$$

$$16y^2 - 10y + 1 \geq 0$$

The roots of the quadratic above will be $\frac{10 \pm 6}{32} = \frac{1}{2}$ or $\frac{1}{8}$

Since the coefficient of y^2 is positive, the quadratic will be less than zero in the range $\left(\frac{1}{8}, \frac{1}{2}\right)$

The quadratic will be greater than or equal to zero otherwise.

Therefore, the domain the quadratic, or possible values of y , which is the range of $f(x)$, will be,

$$\left(-\infty, \frac{1}{8}\right] \cup \left[\frac{1}{2}, \infty\right)$$

Option C is the correct answer.

 [VIEW SOLUTION](#)

27. A

We check where $f(x) = \frac{x}{2x-1}$ or $g(x) = \frac{x}{x-1}$, or their compositions, become undefined.

First, $f(x)$ is undefined at $x = \frac{1}{2}$ and $g(x)$ is undefined at $x = 1$.

$$\text{Next, for } f(g(x)) = \frac{x}{x+1}$$

Since, the denominator can't be zero, $x = -1$ must also be excluded.

$$\text{For } g(f(x)) = \frac{x}{1-x}$$

$\Rightarrow x = 1$ is not possible, which is already excluded.

So the values at which $h(x) = f(g(x)) + g(f(x))$ is undefined are $-1, \frac{1}{2}, 1$

VIDEO SOLUTION

28. A

$$f(3) = \frac{15+2}{9-5} = \frac{17}{4}$$

$$f(f(3)) = \frac{5 \cdot 17/4 + 2}{3 \cdot 17/4 - 5} = \frac{93/4}{31/4} = \frac{93}{31} = 3$$

$$g(f(f(3))) = 3^2 - 3 \cdot 2 - 1 = 2$$

VIDEO SOLUTION

29. B

$$2 \cos(x(x+1)) = 2^x + 2^{-x}$$

The maximum value of LHS is 2 when $\cos(x(x+1))$ is 1 and the minimum value of RHS is 2 using $AM \geq GM$

Hence LHS and RHS can only be equal when both sides are 2. For LHS, $\cos x(x+1)=1 \Rightarrow x(x+1)=0 \Rightarrow x=0, -1$

For RHS minimum value, $x=0$

Hence only one solution $x=0$

VIDEO SOLUTION

30. D

$$y = \frac{1}{2 + \frac{1}{3 + \frac{1}{2 + \frac{1}{3 + \dots}}}}$$

which is equal to $y = \frac{1}{2 + \frac{1}{3+y}}$ solving we get

$$y = \frac{3+y}{7+2y} \text{ we get}$$

$$2y^2 + 6y - 3 = 0.$$

Solution of this equation is $\frac{\sqrt{15}-3}{2}$.

VIDEO SOLUTION

100+ CAT VARC Questions



31. B

When $x < r$

$$f(x) = r$$

$$f(x) = f(f(x))$$

$$r = f(r)$$

$$r = 2r - r$$

$$r = r$$

When $x \geq r$

$$f(x) = 2x - r$$

$$f(x) = f(f(x))$$

$$2x - r = f(2x - r)$$

$$2x-r = 2(2x-r) - r$$

$$2x-r = 4x-3r$$

$$\text{or, } x=r$$

Therefore $x \leq r$

[VIDEO SOLUTION](#)

32. 10

Assuming m is even, then $8f(m+1)-f(m)=2$

$m+1$ will be odd

$$\text{So, } 8(m+1+3)-m(m+1)=2$$

$$\Rightarrow 8m+32-m^2-m=2$$

$$\Rightarrow m^2 - 7m - 30 = 0$$

$$\Rightarrow m=10, -3$$

Rejecting the negative value, we get $m=10$

Assuming m is odd, $m+1$ will be even.

$$\text{then, } 8(m+1)(m+2)-m-3=2$$

$$\Rightarrow 8(m^2 + 3m + 2)-m-3=2$$

$$\Rightarrow 8m^2 + 23m + 11 = 0$$

Solving this, $m = -2.26$ and -0.60

Hence, the value of m is not integral. Hence this case will be rejected.

[VIDEO SOLUTION](#)

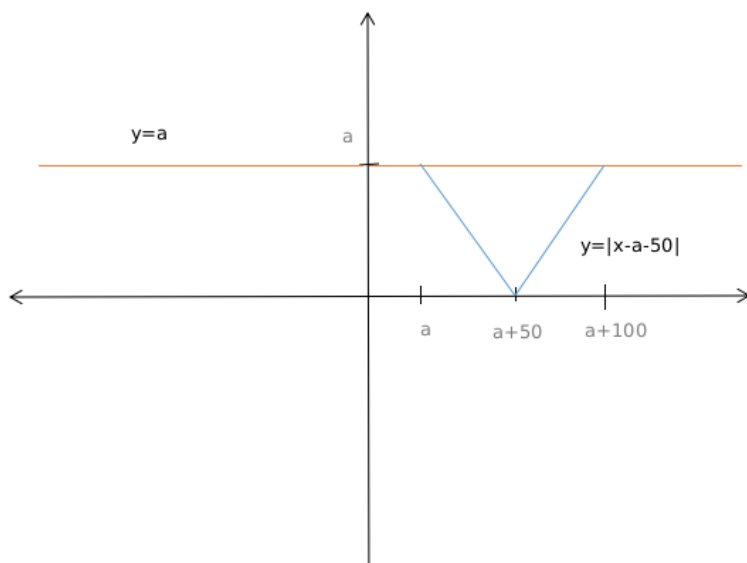
33. C

$$x \geq a, \text{ so } |x-a| = x-a$$

$$x < 100, \text{ so } |x-100| = 100-x$$

$$f(x) = (x-a) + (100-x) + |x-a-50| = 100$$

$$\text{or, } |x-a-50| = a$$



From the graph we can see that when $x=a$ then

$$|x-a-50|=a$$

$$\text{or, } a = 50$$

Similarly when $x=a+100$

$$|x-a-50|=a$$

$$\text{or, } a = 50$$

So value of a is 50 when $f(x)$ is 100.

 VIDEO SOLUTION

34.44

It is given,

$$\sum_{n=1}^N \left[\frac{1}{5} + \frac{n}{25} \right] = 25$$

$$\sum_{n=1}^N \left[\frac{5+n}{25} \right] = 25$$

For $n = 1$ to $n = 19$, value of function is zero.

For $n = 20$ to $n = 44$, value of function will be 1.

$$44 = 20 + n - 1$$

$n = 25$ which is equal to given value.

This implies $N = 44$

 VIDEO SOLUTION

35.20

The minimum value of the function will occur when the expressions inside the function are equal.

$$\text{So, } 5x = 52 - 2x^2$$

$$\text{or, } 2x^2 + 5x - 52 = 0$$

$$\text{On solving, we get } x = 4 \text{ or } -\frac{13}{2}$$

But, it is given that x is a positive number.

$$\text{So, } x = 4$$

$$\text{And the minimum value} = 5 \cdot 4 = 20$$

Hence, 20 is the correct answer.

 VIDEO SOLUTION

CAT Expected Questions PDF

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36.C

$$f[f(g(1)) + g(f(1))]$$

$$= f[f(2^1) + g(1^2)]$$

$$= f[f(2) + g(1)]$$

$$= f[2^2 + 2^1]$$

$$= f(6)$$

$$= 6^2 = 36$$

 VIDEO SOLUTION

37. D

Now we have :

$$f(g(x)) - 3x$$

so we get $f(x+3)-3x$

$$= (x+3)^2 - 7(x+3) - 3x$$

$$= x^2 - 4x - 12$$

$$\text{Now minimum value of expression} = -\frac{D}{4a} \frac{(4ac-b^2)}{4a}$$

$$\text{We get } - (16+48)/4$$

$$= -16$$

 VIDEO SOLUTION

38. A

According to given conditions we get $f(2)=f(1)/3$, then $f(3)=f(1)/6$, then $f(4)=f(1)/10$, then $f(5)=f(1)/15$.

We can see the pattern here that the denominator goes on increasing from 3,3+3,6+4,10+5,15+6,... so for the $f(9)$ the denominator will be same as $15+6+7+8+9=45$.

$$\text{So } f(9)=3600/45 = 80$$

 VIDEO SOLUTION

39. 12

$$\text{Given, } f(mn) = f(m)f(n)$$

$$\text{when } m= n= 1, f(1) = f(1)*f(1) \implies f(1) = 1$$

$$\text{when } m=1, n= 2, f(2) = f(1)*f(2) \implies f(1) = 1$$

$$\text{when } m=n= 2, f(4) = f(2)*f(2) \implies f(4) = [f(2)]^2$$

$$\text{Similarly } f(8) = f(4)*f(2) = [f(2)]^3$$

$$f(24) = 54$$

$$[f(2)]^3 * [f(3)] = 3^3 * 2$$

On comparing LHS and RHS, we get

$$f(2) = 3 \text{ and } f(3) = 2$$

Now we have to find the value of $f(18)$

$$f(18) = [f(2)] * [f(3)]^2$$

$$= 3*4=12$$

 VIDEO SOLUTION

40. B

$$f(1) = 1-4+p = p-3$$

$$f(0) = p$$

Since they are of opposite signs, $p(p-3) < 0$

$$\implies 0 < p < 3$$

 VIDEO SOLUTION

 **CAT Important Topics** 

41. B

$$f(1)^2 = f(1) \Rightarrow f(1) = 1$$

$$f(2) \cdot f(1/2) = f(1) \Rightarrow 4x = 1$$

$$\text{So, } f(1/2) = 1/4$$

 VIDEO SOLUTION

42. B

$$f(3) = 9a + 3b + c = 0 \quad f(5) = 25a + 5b + c$$

$$f(2) = 4a + 2b + c$$

$$f(5) = -3f(2) \Rightarrow 25a + 5b + c = -12a - 6b - 3c$$

$$\Rightarrow 37a + 11b + 4c = 0 \rightarrow (1)$$

$$4(9a + 3b + c) = 36a + 12b + 4c = 0 \rightarrow (2)$$

$$\text{From (1) and (2), } a - b = 0 \Rightarrow a = b$$

$$\Rightarrow c = -12a$$

$$\text{The equation is, therefore, } ax^2 + ax - 12a = 0 \Rightarrow x^2 + x - 12 = 0$$

$$\Rightarrow -4 \text{ is a root of the equation.}$$

 VIDEO SOLUTION

43. E

$$f(3) = 9a + 3b + c = 0 \quad f(5) = 25a + 5b + c$$

$$f(2) = 4a + 2b + c$$

$$f(5) = -3f(2) \Rightarrow 25a + 5b + c = -12a - 6b - 3c$$

$$\Rightarrow 37a + 11b + 4c = 0 \rightarrow (1)$$

$$4(9a + 3b + c) = 36a + 12b + 4c = 0 \rightarrow (2)$$

$$\text{From (1) and (2), } a - b = 0 \Rightarrow a = b$$

$$\Rightarrow c = -12a$$

$$\text{The equation is, therefore, } ax^2 + ax - 12a = 0 \Rightarrow x^2 + x - 12 = 0$$

$$a + b + c = a + a - 12a = -10a.$$

But the value of a is not given. Therefore, the value cannot be determined.

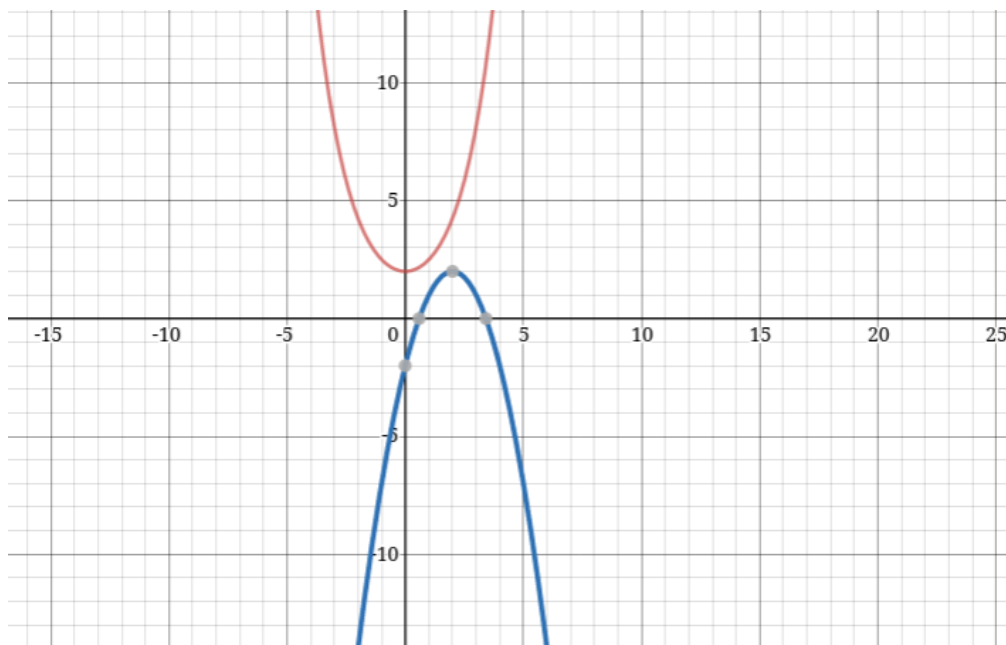
 VIDEO SOLUTION

44. D

Equate the 2 equations we get value of $x = 1$ and -1 . Also we notice that there is intersection at $x=0$. hence D

 VIDEO SOLUTION

45. D



The graphs of $2^x + 2^{-x}$ and $2 - (x - 2)^2$ never intersect. So, number of solutions = 0.

Alternate method:

We notice that the minimum value of the term in the LHS will be greater than or equal to 2 {at $x=0$; LHS = 2}. However, the term in the RHS is less than or equal to 2 {at $x=2$; RHS = 2}. The values of x at which both the sides become 2 are distinct; hence, there are zero real-valued solutions to the above equation.

[VIDEO SOLUTION](#)



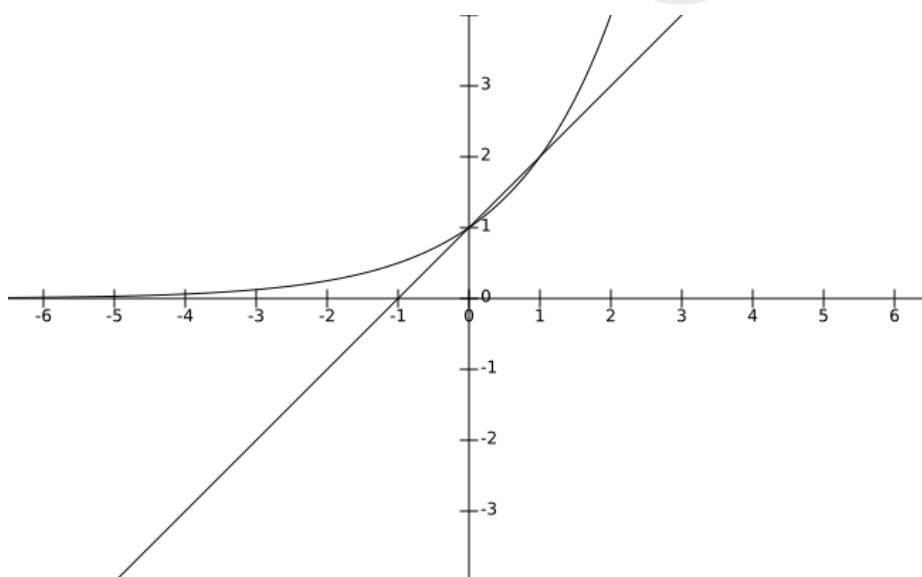
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46. C

$2^x - x - 1 = 0$ for this equation only 0 and 1 i.e 2 non-negative solutions are possible. Or we can plot the graph of 2^x and $x+1$ and determine the number of points of intersection and hence the solution.



[VIDEO SOLUTION](#)

47. 3

$$f(x+y) = f(x) f(y)$$

$$\text{Hence, } f(2)=f(1+1)=f(1)*f(1)=2*2=4$$

$$f(3)=f(2+1)=f(2)*f(1)=4*2=8$$

$$f(4)=f(3+1)=f(3)*f(1)=8*2=16$$

$$\dots\dots \Rightarrow f(x)=2^x$$

$$\text{Now, } f(a+1) + f(a+2) + \dots + f(a+n) = 16(2^n - 1)$$

On putting $n=1$ in the equation we get, $f(a+1)=16 \Rightarrow f(a)*f(1)=16$ (It is given that $f(x+y) = f(x)f(y)$)

$$\Rightarrow 2^a * 2 = 16$$

$$\Rightarrow a=3$$

[▶ VIDEO SOLUTION](#)

48. B

$$f(x) = x^2 + ax + b$$

$$f(x+1) = x^2 + 2x + 1 + ax + a + b$$

$$f(x-1) = x^2 - 2x + 1 + ax - a + b$$

$$g(x) = f(x+1) - f(x-1) = 4x + 2a$$

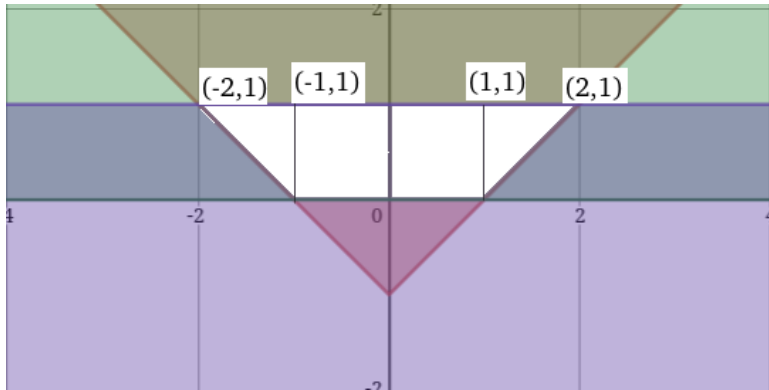
$$\text{Now } g(20) = 72 \text{ from this we get } a = -4; f(x) = x^2 - 4x + b$$

For this expression to be greater than zero it has to be a perfect square which is possible for $b \geq 4$

Hence the smallest value of 'b' is 4.

[▶ VIDEO SOLUTION](#)

49. 3



The area of the region contained by the lines $|x| - y \leq 1, y \geq 0$ and $y \leq 1$ is the white region.

$$\text{Total area} = \text{Area of rectangle} + 2 * \text{Area of triangle} = 2 + \left(\frac{1}{2} \times 2 \times 1\right) = 3$$

Hence, 3 is the correct answer.

[▶ VIDEO SOLUTION](#)

50. D

$$x^2 - 2|x| + |a - 2| = 0$$

where $x \geq 0$ and $x \leq 2$

$x^2 - 2x + a - 2 = 0$ Using quadratic equation we have $x = 1 + \sqrt{3 - a}$ and $x = 1 - \sqrt{3 - a}$ Only two integer values are possible

$a=2$ and $a=3$. So corresponding "x" values are $x=1$ and $a=3$, $x=2$ and $a=2$, $x=0$ and $a=2$

where $x \geq 0$ and $x < 2$

Applying the above process we get $x=1$ and $a=1$

where $x<0$ and $x\geq 2$ we get $a=3$ and $x=-1$, $a=2$ and $x=-2$

where $x<0$ and $x<2$ we get $a=1$ and $x=-1$

Hence there are total 7 values possible

 VIDEO SOLUTION



51. B

Let $x(1)$ be the least number and $x(10)$ be the largest number. Now from the condition given in the question, we can say that

$$x(2)+x(3)+x(4)+\dots+x(10)=47*9=423\dots\dots(1)$$

$$\text{Similarly } x(1)+x(2)+x(3)+x(4)+\dots+x(9)=42*9=378\dots\dots(2)$$

Subtracting both the equations we get $x(10)-x(1)=45$

Now, the sum of the 10 observations from equation (1) is $423+x(1)$

Now the minimum value of $x(10)$ will be 47 and the minimum value of $x(1)$ will be 2. Hence minimum average $425/10=42.5$

Maximum value of $x(1)$ is 42. Hence maximum average will be $465/10=46.5$

Hence difference in average will be $46.5-42.5=4$ which is the correct answer

 VIDEO SOLUTION

52. C

The given function is equivalent to $f(x) = a^x$

Given, $f(5) = 4$

$$\Rightarrow a^5 = 4 \Rightarrow a = 4^{\frac{1}{5}}$$

$$\Rightarrow f(x) = 4^{\frac{x}{5}}$$

$$f(10) - f(-10) = 16 - 1/16 = 15.9375$$

 VIDEO SOLUTION

53. 24

$$f_1(x) = x^2 + 11x + n \text{ and } f_2(x) = x$$

$$f_1(x) = f_2(x)$$

$$\Rightarrow x^2 + 11x + n = x$$

$$\Rightarrow x^2 + 10x + n = 0$$

$\Rightarrow$ For this equation to have distinct real roots, $b^2-4ac>0$

$$10^2 > 4n$$

$$\Rightarrow n < 100/4$$

$$\Rightarrow n < 25$$

Thus, largest integral value that n can take is 24.

 VIDEO SOLUTION

54. 3

Given that $f(3x + 2y, 2x - 5y) = 19x$.

Let us assume the function $f(a,b)$ is a linear combination of a and b .

$$\Rightarrow f(3x+2y, 2x-5y) = m(3x+2y) + n(2x-5y) = 19x$$

$$\Rightarrow 3m + 2n = 19 \text{ and } 2m - 5n = 0$$

Solving we get $m = 5$ and $n = 2$

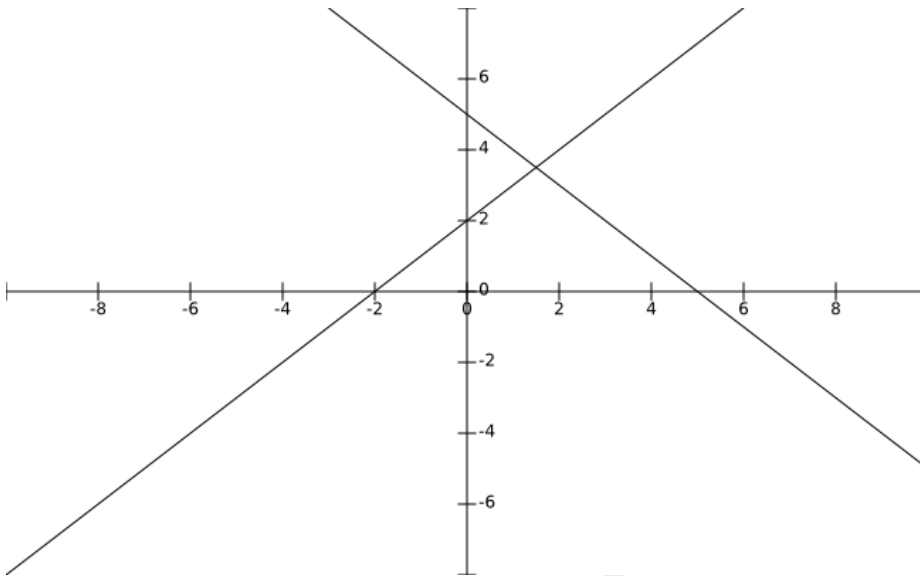
$$\Rightarrow f(a,b) = 5a+2b$$

$$\Rightarrow f(x,2x) = 5x + 2(2x) = 9x = 27 \Rightarrow x = 3.$$

[VIDEO SOLUTION](#)

55. D

Smallest possible value would be at $5-x = x+2$ i.e. $x = 1.5$ as shown



Substituting we get smallest value as 3.5.

[VIDEO SOLUTION](#)

A banner for 'CAT Online Coaching' with the tagline 'your road to IIM stars here'. It features a photo of two people on the left and a person working on a laptop on the right. The 'cracku' logo is in the top right corner.

56. D

According to given condition we have , $g(x+1) = -g(x-1) + g(x)$

Putting $x=x+1$ we get $g(x+2) = g(x+1) - g(x) = -g(x-1)$

Putting $x=x+2$ we get $g(x+3)=-g(x)$

Similarly $g(x+4)=-g(x+1)$, $g(x+5)=-g(x+2)=-g(x+1) + g(x)$ and $g(x+6) = g(x+1)-g(x+2)=g(x)$.

So $p=6$.

[VIDEO SOLUTION](#)

57. D

$f(x) = ax^2 - b|x|$. When $x = 0$, $f(x) = 0$

When $a > 0$ and $b < 0$,

For $x > 0$, $f(x) = ax^2 - bx$, will be greater than 0 as $ax^2 > 0$ and $bx < 0$ as b is negative and x is positive.

For $x < 0$, $f(x) = ax^2 + bx$ will again be greater than 0 as $ax^2 > 0$ and $bx > 0$ as both b and x are negative.

Therefore, the function $f(x)$ is positive when $x < 0$ and when $x > 0$ but becomes 0 when $x = 0$.

Therefore, for $a > 0$ and $b < 0$, $f(x)$ will attain its minimum value at $x = 0$.

 VIDEO SOLUTION

58.1

$$f(1 * 1) = f(1)f(1)$$

$$\Rightarrow f(1) = f(1)f(1)$$

$$\Rightarrow f(1) = 0 \text{ or } f(1) = 1$$

Hence maximum value of $f(1)$ is 1

 VIDEO SOLUTION

59. B

$$f(x) = |x - 2| + |2.5 - x| + |3.6 - x|$$

For x belonging to $(-\infty, 2)$, $f(x) = 2 - x + 2.5 - x + 3.6 - x = 8.1 - 3x$

This attains the minimum value at $x = 2$. Value = 2.1

For x belonging to $(2, 2.5)$, $f(x) = x - 2 + 2.5 - x + 3.6 - x = 4.1 - x$

Attains the minimum value at $x = 2.5$. Value = 1.6

For x belonging to $(2.5, 3.6)$, $f(x) = x - 2 + x - 2.5 + 3.6 - x = x - 0.9$

Attains the minimum at $x = 2.5$, value = 1.6

For $x > 3.6$, $f(x) = x - 2 + x - 2.5 + x - 3.6 = 3x - 8.1$

Attains the minimum at $x = 3.6$, value = 2.7

So, min value of the function is 1.6 at $x = 2.5$

 VIDEO SOLUTION

60. D

Consider different values of x and y :

$x = -1.5$ and $y = -1.5$; $x = 1.5$ and $y = -1.5$; $x = -1.5$ and $y = 1.5$; $x = 1.5$ and $y = 1.5$.

For these possibilities, options A, B and C gets satisfied, but it is impossible to find any two positive real numbers x and y for which $L(x, y) > R(x, y)$.

 VIEW SOLUTION



61. C

Using given condition we find $a_2 = 5$ and $a_3 = 21$ and so on.

We see that the numbers are of form $3^n - (2 * n)$

So for 100 we have $3^{100} - 200$

 VIEW SOLUTION

62. C

If the moduli are removed, the equations formed are

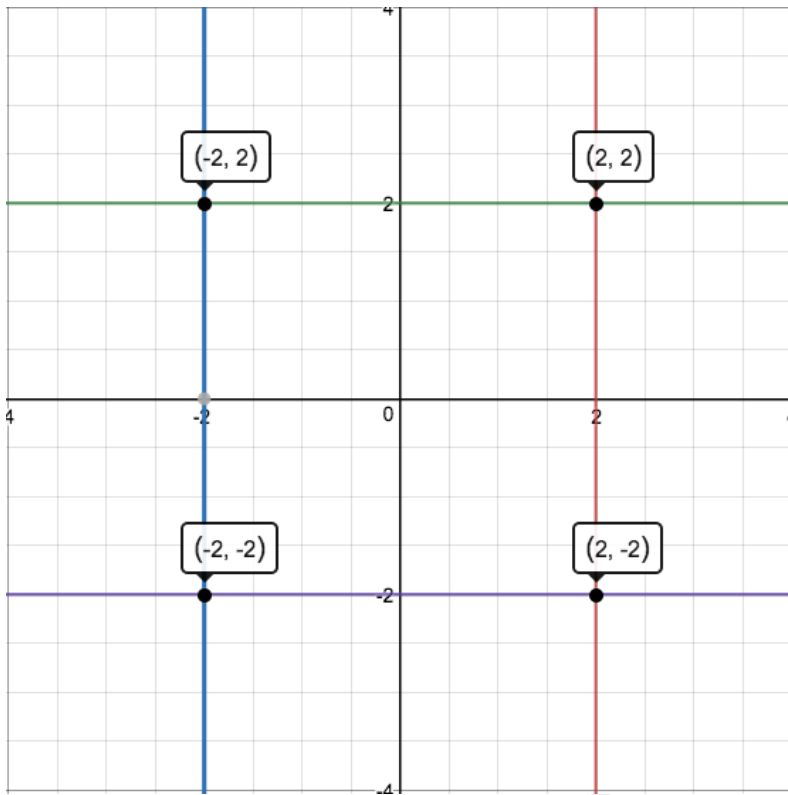
$$x+y+x-y = 4 \Rightarrow x=2$$

$$x+y-x+y = 4 \Rightarrow y=2$$

$$-x-y+x-y = 4 \Rightarrow y=-2$$

$$-x-y-x+y = 4 \Rightarrow x=-2$$

The area enclosed by these equations is a square with vertices at (2,2), (-2,2), (-2,-2), (2,-2) as shown in figure.



The required area = $4 \times 4 = 16$

[VIEW SOLUTION](#)

63. D

Let 'r' be the root of the function. It follows that $f(r) = 0$. We can represent this as $f(r) = f\{5 - (5 - r)\}$

Based on the relation: $f(5 - x) = f(5 + x)$; $f(r) = f\{5 - (5 - r)\} = f\{5 + (5 - r)\}$

$$\therefore f(r) = f(10 - r)$$

Thus, every root 'r' is associated with another root '(10-r)' [these form a pair]. For even distinct roots, in this case four, let us assume the roots to be as follows: $r_1, (10 - r_1), r_2, (10 - r_2)$

$$\text{The sum of these roots} = r_1 + (10 - r_1) + r_2 + (10 - r_2) = 20$$

Hence, Option D is the correct answer.

[VIDEO SOLUTION](#)

64. B

$f(x) \geq 0$ for all real numbers x , so $D \leq 0$

Since $f(2)=0$ therefore $x=2$ is a root of $f(x)$

Since the discriminant of $f(x)$ is less than equal to 0 and 2 is a root so we can conclude that $D=0$

$$\text{Therefore } f(x) = a(x - 2)^2$$

$$f(4)=6$$

$$\text{or, } 6 = a(x-2)^2$$

$$a = 3/2$$

$$f(-2) = -\frac{3}{2}(-4)^2 = -24$$

 VIDEO SOLUTION

65. A

Looking at the additional information about the prime numbers should make one realise that they are the key to solving the question.

$f(16000)$ can be written as $f(2^8 \times 5^4)$

Now, we can try to find these individual values:

For any prime p : $f(p)=1$

$$f(p^2) = f(p)f(p) + f(p) + f(p) = 1 + 1 + 1 = 3$$

$$f(p^3) = f(p^2)f(p) + f(p^2) + f(p) = 3 + 3 + 1 = 7$$

This way, we can find the function output for any prime number raised to a power.

We can see that each new exponent is twice the previous output +1, solving this way till prime raised to power 8

$$f(p^4) = 7 + 7 + 1 = 15$$

$$f(p^5) = 15 + 15 + 1 = 31$$

$$f(p^6) = 31 + 31 + 1 = 63$$

$$f(p^7) = 63 + 63 + 1 = 127$$

$$f(p^8) = 127 + 127 + 1 = 255$$

Using these values in the original expression of $f(2^8 \times 5^4) = f(2^8)f(5^4) + f(2^8) + f(5^4)$ we get

$$f(2^8 \times 5^4) = (255 \times 15) + 255 + 15 = 4095$$

Therefore, Option A is the correct answer.

 VIDEO SOLUTION

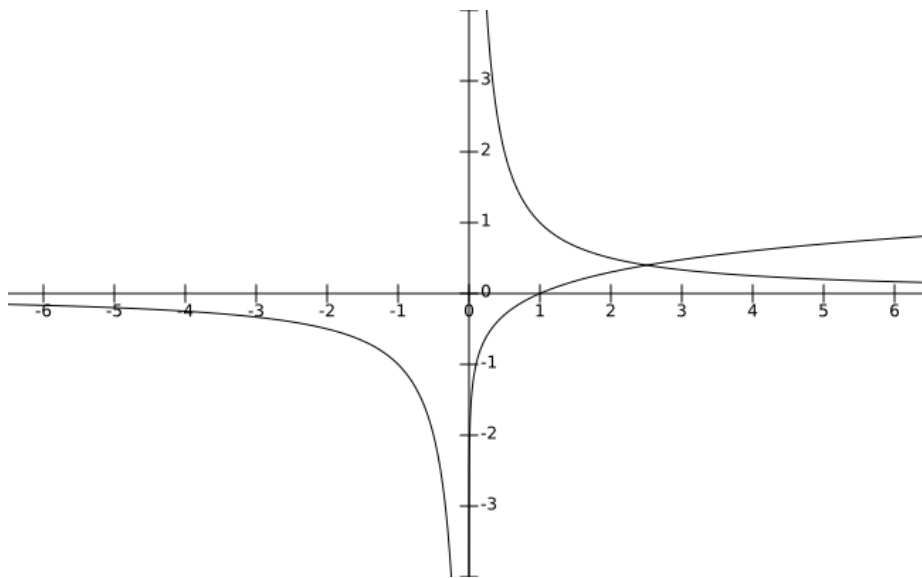


CAT Syllabus PDF



66. B

Graph of $\log x$ goes on increasing in 1st quadrant and graph of $1/x$ goes on decreasing with both intersecting only once



[VIDEO SOLUTION](#)

67. **A**

When both x and y are less than -1 , $g(x,y) > 2$ and $f(x,y) > 4$ and $f(x,y) = g(x,y)^2$
 So, $f(x,y) > g(x,y)$

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68. **D**

$f(x,y)$ is always non-negative because $(x+y)^2$ is always positive

$g(x,y) = (x+y)^2$, if $\sqrt{(x+y)}$ is real = Always positive

$g(x,y) = -(x+y)$ otherwise = Always positive because this happens when $(x+y) < 0$ and $-(x+y)$ is always greater than zero.

$f(x,y) + g(x,y)$ = Always positive

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69. **D**

For n , the number of elements in set S is nC_2 .

Lets say the 2 friends are (x,a) and (y,a)

These two friends have 3 numbers in total and 1 common element (say a) (as both elements cannot be exactly same)

They have 2 non common elements (x,y)

The number of common friends is formed by the non-common elements of the friends (x,y) + the number of elements in the set which have the common element other than the two friends (a,c) , (a,d) and so on $= 1 + (n-1-2) = n-2$.

For the example in question, if the friends are $(1,2)$ and $(1,3)$, then common friends are $(2,3)$ and all other elements with 1

All elements with 1 $= n-1 = 3$ which are $(1,2)$ $(1,3)$ $(1,4)$ excluding the friends $(1,2)$ and $(1,3)$ only 1 other friend is common. Hence it is $1+(n-1)-2=n-2$

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70. **D**

Any ordered pair has 2 elements $\Rightarrow$ There are $n-2$ elements that are not present in the ordered pair.

The number of enemies of any ordered pair is all the ordered pairs in the set formed using the numbers other than these two elements = ${}^{n-2}C_2 = 1/2 * (n^2 - 5n + 6)$.

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71. **C**

$$F(0) = 1 ; F(1)(0) = -1$$

$$F(1) = 0 ; F(1)(-1) = 0$$

$$F(2) = -1 ; F(1)(-2) = 1$$

$$\Rightarrow F(1)(x) = -F(-x).$$

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72. **B**

The value of $F(x)$ for $x < 0$ is the same as the value of $F(1)(x)$ for $x > 0$.

$$\text{So, } F(1)(x) = F(-x)$$

Option b) is the correct answer.

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73. **B**

The value of $F(x)$ for $x > 0$ is the same as the value of $F(1)(x)$ for $x < 0$.

$$\text{So, } F(1)(x) = F(-x)$$

Option b) is the correct answer.

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74. **D**

The correct relation between the two is: $F(x) = |F(1)(x)|$

So, all the three options a), b) and c) can be ruled out. Option d) is the correct answer.

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75. **C**

In the given graph, the value of y is constant irrespective of the value of x . Hence value of y would be same for a particular x and $-x$.

$$\text{Hence } f(x) = f(-x)$$

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76. **D**

In this graph, $f(-1) = 1$ and $f(1) = 2$ and since the two lines pass origin, their values increase linearly.

$$3f(x) = 6f(-x)$$

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77. **B**

In this graph, we can see that the graph on LHS of y axis is symmetrical to the graph on RHS of y axis about origin.

$$\text{Hence } f(x) = -f(-x)$$

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78. **C**

$$f(2) = 1/3$$

$$f^2(2) = f(1/3) = 3/4$$

$$f^3(2) = f(3/4) = 4/7$$

$$f^4(2) = f(4/7) = 7/11$$

$$f^5(2) = f(7/11) = 11/18$$

$$\text{So, product} = 1/18$$

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79. **B**

$$f(-2) = -1$$

$$f^2(-2) = f(f(-2)) = f(-1) = 0$$

$$f^3(-2) = f(f(-2)) = f(0) = 1$$

$$\text{So, sum} = 0$$

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80. **A**

WALKX(-x) and WALKY(y) are the commands to be used for the robot to reach origin in any order.

Hence, the answer is 2.

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CAT Percentile Predictor



81. **C**

Before, the third instruction, the point on which the robot is present is (6,2).

Before, the second instruction, the point on which the robot is present is (4,2).

Hence, the values of x and y are 4 and 2 respectively.

[VIEW SOLUTION](#)

82. **B**

Given expression can be reduced to:

$$\begin{aligned}
 & S\left[M\left(\frac{a+b}{2}\right), \left(\frac{a+b}{2}\right)\right], M\left(\left(\frac{a-b}{2}\right), \left(\frac{a-b}{2}\right)\right] \\
 &= S\left[\left(\frac{a+b}{2}\right)^2, \left(\frac{a-b}{2}\right)^2\right] \\
 &= \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2 \\
 &= ab
 \end{aligned}$$

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83. D

Given expression can be reduced to

$$M(x(xy+y-x), (y+x))$$

$$\text{Or } x(x+y)(xy+y-x)$$

After putting value of x and y, expression will reduce to a value = 70.

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84. D

From the given functions we can make out that function h and m give max value, function n and j give min value, function f and g give middle value. From this equation $(f(x, y, z) + h(x, y, z) - g(x, y, z)) / j(x, y, z)$, numerator is always max value and denominator is min value. So this will always be greater than 1.

Suppose $x > y > z$

$$f(x, y, z) = y$$

$$g(x, y, z) = y$$

$$h(x, y, z) = x$$

$$j(x, y, z) = z$$

$$\text{Option d} = x/z > 1$$

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85. B

From the given functions we can make out that function h and m give max value, function n and j give min value, function f and g give middle value. So in option B, j cancels out n and h cancels out m. So the denominator becomes 0 and value is indeterminable.

Suppose $x > y > z$

$$f(x, y, z) = y$$

$$g(x, y, z) = y$$

$$h(x, y, z) = x$$

$$j(x, y, z) = z$$

$$m(x, y, z) = x$$

$$n(x, y, z) = z$$

The denominator of the second option becomes 0, hence making it indeterminate.

[VIEW SOLUTION](#)



86. A

From the given functions we can make out that function h and m give max value, function n and j give min value, function f and g give middle value. So according to equation $(f(x, y, z) - m(x, y, z)) / (g(x, y, z) - h(x, y, z))$, value of numerator and denominator is equal and hence ratio is equal to 1.

Suppose $x > y > z$

$$f(x, y, z) = y$$

$$g(x, y, z) = y$$

$$h(x, y, z) = x$$

$$j(x, y, z) = z$$

$$\text{Option a} = (y - x) / (y - x) = 1$$

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87. D

$$f(x, y) = |x + y|$$

$$F(f(x, y)) = -f(x, y) = -|x + y|$$

$$G(f(x, y)) = -F(f(x, y)) = |x + y|$$

Option A: $F(f(x, y)) \cdot G(f(x, y)) = -F(f(x, y)) \cdot G(f(x, y)) \Rightarrow \text{LHS} = -|x + y|^2$, $\text{RHS} = |x + y|^2$ Hence false.

Option B: $F(f(x, y)) \cdot G(f(x, y)) > -F(f(x, y)) \cdot G(f(x, y))$, Since, LHS is smaller than RHS. False

Option C: $F(f(x, y)) \cdot G(f(x, y)) \neq G(f(x, y)) \cdot F(f(x, y))$, Here $\text{LHS} = \text{RHS}$. Hence false.

Option D:

$$\Rightarrow G(f(x, y)) + F(f(x, y)) = 0$$

$$f(x, y) = f(-x, -y)$$

$$\Rightarrow G(f(x, y)) + F(f(x, y)) + f(x, y) = f(-x, -y)$$

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88. C

$$F(f(x, y)) = -f(x, y)$$

$$G(f(x, y)) = -F(f(x, y)) = f(x, y)$$

$$G(f(1, 0)) = 1$$

$$F(f(1, 2)) = -3$$

$$G(f(1, 2)) = 3$$

$$f(F(f(1, 2)), G(f(1, 2))) = 0$$

$$f(G(f(1, 0)), f(F(f(1, 2))), G(f(1, 2))) = 1 + 0 = 1$$

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89. C

$$F(f(x, -x)) = 0 \text{ and } G(f(x, -x)) = 0$$

$$F(f(x, x)) = -2x \text{ and } G(f(x, x)) = 2x$$

$$F(f(x, x)).G(f(x, x)) = -4x^2$$

$$\log_2 16 = 4$$

$$\Rightarrow \frac{-F(f(x, x)).G(f(x, x))}{\log_2 16} = x^2$$

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90. B

Relation between f_3 and f_1 would be

$$f_3(x) = -f_1(-x).$$

Put $x = -x$ we get

$$f_3(-x) = -f_1(x) \text{ so multiply by } -1 \text{ we get}$$

$$-f_3(-x) = f_1(x).$$

[VIDEO SOLUTION](#)



91. C

Checking for different values of x . Suppose $x = -0.5$ we get

$$f_1(x)f_2(x) = 0 * 0.5 = 0$$

$$f_2(x)f_4(x) = 0.5 * 0 = 0.$$

But $f_2(x)f_3(x)$ is not equal to zero.

Hence two functions are necessarily equal to zero and two products given above are equal to zero.

[VIDEO SOLUTION](#)

92. B

$$a_n + b_n \text{ for even } n = p * b_{n-1} + q * b_{n-1}$$

$$= (p + q) * b_{n-1}$$

$$b_{n-1} = q * a_{n-2} = qp * b_{n-3}$$

$$= q^2 * p * a_{n-4} = q^2 p^2 * b_{n-5}$$

.

.

$$= (qp)^{n/2-1} * b_1 = (qp)^{n/2-1} * q$$

$$\text{So, } a_n + b_n = q(pq)^{(n/2)-1}(p + q)$$

 [VIEW SOLUTION](#)

93. D

$$a_n + b_n \text{ (n is odd)} = p^{\frac{n+1}{2}} * q^{\frac{n-1}{2}} + p^{\frac{n-1}{2}} * q^{\frac{n+1}{2}} = (p + q)pq^{\frac{n-1}{2}}$$

Substituting the values of p and q we get

$$a_n + b_n = \left(\frac{2}{9}\right)^{\frac{n-1}{2}}$$

Now substitute the values of n and check.

We can see that the lowest value of n for which

$$a_n + b_n < .01 \text{ is } 9$$

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